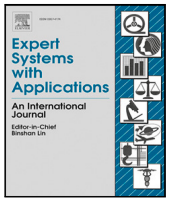
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Masked face age and gender identification using CAFFE-modified MobileNetV2 on photo and real-time video images by transfer learning and deep learning techniques

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ABSTRACT

One of the most challenging factors related to masked face age and gender identification (MFAGI) is developing a technique to quickly carry out identification and maintain accuracy without needing people to remove their masks. The eyes, forehead and important frontal features including the ears were the main focus of this investigation. In this study, we evaluated and compared ten (10) different MFAGI techniques, which include traditional methods such as MobileNet, MobileNetV2, ResNet101, DenseNet169, DenseNet201, EfficientNetB0, ResNet152V2, InceptionV3, Xception and our one proposed approach. These 10 models were used to classify masked faces of people among sixteen (16) classes. Combining the convolutional architecture of fast feature embedding (CAFFE) with the modified MobileNetV2 (MNV2) model developed a novel method known as the "CMNV2" model. To achieve the goal of model performance and classification accuracy for MFAGI of people's faces, this work introduces a unique CMNV2 model by developing a pre-trained model with eight (8) different layers. Using deep learning (DL) and a transfer learning approach, we developed a unique novel approach to image feature extraction that outperforms the nine (9) traditional methods for masked face classification (MFC), masked face age identification (MFAI) and masked face gender identification (MFGI). This approach developed a Python model that was trained and evaluated on 12,160 masked face images. This study demonstrates that an efficient CMNV2 method can be achieved by combining transfer learning with deep neural network (DNN) development. This proposed CMNV2 model achieved 96.54% accuracy and 3.46% error rate with less parameters and computational time exceeding the other 9 different traditional masked identification methods. It also shows that our proposed CMNV2 strategy is more successful than the other 9 models and can identify masked people's faces age and gender in photo and real-time video images. Overall, the results of this research indicate that DL methods have the capability to improve the accuracy and speed with which masked face classification and identification is performed. The artificially masked faces of images dataset created by us utilized in this work is publicly made available on the Kaggle website https://www.kaggle.com/datasets/banilkumar20phd7071/masked-face-age-and-gender-identify-artificial-masking.

1. Introduction

Facial identification, age and gender categorization and facial gesture recognition systems are vital applications in image processing and deep learning. Surveillance, biometrics, security, image labeling and identity verification systems rely heavily on accurate face identification. Image processing importance has grown recently as computers can distinguish between male and female categories based on unique personality characteristics. Age and gender classification are critical data-driven issues applicable in various situations. Researchers estimate age alongside gender for more reliable results. Gender determination involves analyzing geometric differences in primary features. Wrinkles, indicative of age, contribute to accurate age estimations. Two independent artificial neural networks categorize age and gender. The dataset for this method includes images of people of different ages and genders, featuring various facial features, emotions, camera angles, backgrounds and lighting.

Features-based geometry can identify a related face even when the eyes, nose and mouth are unique. Most of the remaining data is geometrical and has a poor resolution. The purpose is to identify and size eyes, mouth, nose, face shape, cheek and chin (Brunelli & Poggio, 1993; Yamaguchi, Hirukawa, & Kanazawa, 2013). Individual

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authentication and identity research is expanding to include physiological features such as the face, voice, lip motions, hand geometry, odor, stride, eyes, retina and fingerprint (Ramesha, Raja, Venugopal, & Patnaik, 2010). Commercial applications like video surveillance and image search engines make use of this unique set of personal and behavioral data to automatically classify face age and gender.

A neural network categorizes images by gender and age using visible changes in men's and women's facial shape and texture as they age (Kalansuriya & Dharmaratne, 2013). Identifying individuals can be challenging when comparing photos of different ages. Age-invariant face identification is important for locating missing people, home security and passport services. Predictability in aging is essential for large-scale applications (Yadav, Singh, Vatsa, & Noore, 2014).

Several companies use automated segmentation of persons for surveillance, security and targeted marketing. Face recognition systems need face image normalization before feature extraction due to localization problems and lighting (Azarmehr, Laganiere, Lee, Xu, & Laroche, 2015). Increased social media and internet use have led to increased applications for automated age and gender categorization (Levi & Hassner, 2015). Face detection using a deep convolutional neural network (DCNN) can estimate ages. Using a Gaussian loss-trained three-layer neural network regression technique generates more accurate age estimates than standard regression methods (Ranjan et al., 2015).

Age and gender predictions are made using CNN and residual networks of residual networks (RoR). This method outperforms CNNs in age and gender categorization optimization. Two key methods that monitor age group features improve age estimates. Pre-training the RoR model on ImageNet and fine-tuning it on the image dataset to learn face image features improves efficiency and reduces overfitting risk (Zhang, Gao, et al., 2017).

Computer vision and pattern recognition researchers investigate face position, expression, lighting, occlusion and other features (Kumar, Singh, & Kumar, 2017). Business systems use log data for statistical analysis and real-time personalized suggestions based on age and gender (Liu, Li, Hu, & Pan, 2017). A feedforward attention technique can identify a face's most important features to improve age and gender identification (Rodríguez, Cucurull, Gonfaus, Roca, & Gonzalez, 2017). Age is quantified using large face picture datasets with correct posterior distributions (Zhang, Liu, et al., 2017). Face, skin and body signals are used to identify gender in many applications, with facial images being the most important (Ghojogh, Shouraki, Mohammadzade, & Iranmehr, 2018).

Transfer learning using a pre-trained model suitable to the issue can outperform humans in gender identity and age prediction. Specific model structures, training methods and pre-trained weights are beneficial. The integration of a categorization system in artificial intelligence (AI) model hierarchy is promising and needs further investigation (Smith & Chen, 2018). A person's age may be determined by their face's texture and contours. Evaluating geometric aspects including wrinkle area, facial angle, eye-to-nose, eye-to-lip, eye-to-chin and left-to-right eye distance improves face recognition algorithms (Mélange & Sable, 2018).

As visual surveillance and human–computer interface technologies progress, age and gender categorization computer vision systems become more important. Highly sociable individuals must visually classify age and gender for social interactions (Dileep & Danti, 2018). Data from post-processed facial images is analyzed across age and gender groups to produce binary masks and estimate wrinkle densities (Rahman, Arefeen, Mridul, Khan, & Subrina, 2020). Automated age and gender categorization, especially using unconstrained mobile images, has grown in importance (Lee, Chan, Chen, & Chen, 2018). Current deep learning approaches have various obstacles including high hardware requirements, a huge amount of model parameters and issues embedding in mobile devices such as drones. Even the most powerful machine-vision systems, however, have limitations. As a result, most studies in this field have shown insufficient measurement accuracy as a result of certain system defects in computer vision modules.

Age and gender prediction performance improves significantly with CNN feature representation training (Wanga & Davies, 2019). Modern biometric and facial identification research focuses on gender and age prediction for better future predictions and a better understanding of people. Deep learning is the main method, with various strategies employed in the process (Al-Azzawi, 2019). Deep CNN frameworks with ear and profile face images are used for age and gender categorization in multitask learning (Yaman, Irem Eyiokur, & Kemal Ekenel, 2019). Studies recommend fundamental changes to visual identification systems and show that balanced datasets may not provide fair predictions (Wang, Zhao, Yatskar, Chang, & Ordonez, 2019).

An overview of the paper's work and a distillation of the innovations and key technical contributions involved:

i. Developed a modified MNV2 model by adding 8 different layers.
ii. A novel CMNV2 proposed approach is designed further using CAFFE and modified MNV2 to train and test for the MFC and MFAGI, which more accurately captured the challenges that occur by mask-wearing people.
iii. We created a dataset of 12,160 images using artificial masking of people's faces with masks for the MFAI and MFGI of 16 classes, these dataset was made publicly accessible on the Kaggle platform.
iv. A Google form has been created that helped us in testing our proposed CMNV2 model against a dataset of photo images that is unseen. A total of 130 participants took part in the survey and filled the Google form willingly.
v. A total of 24 participants belonging to B.Tech students and Ph.D. scholars volunteered to test our novel CMNV2 technique on real-time video image processing and prediction.
vi. This dataset allows safe data transmission for MFGI and MFAI. Researchers and organizations can use our MFAGI dataset without affecting privacy, which addresses ethical issues in image processing using artificial masks and protects individual's privacy.
vii. Our novel technique of CMNV2 structure was developed to increase MFAI and MFGI accuracy with fewer parameters, a lower error rate and less processing time.
viii. Codes were written, algorithms were created and software solutions were developed to meet all aspects of MFAGI.
ix. Testing photo and real-time video image challenges were successfully handled with the assistance of a team of B.Tech. and Ph.D. scholars.
x. This study was created to fill a gap that was identified in MFAGI. A robust technique of the CMNV2 model is used to demonstrate the need for the technology contribution.
xi. The suggested CMNV2 model was evaluated against 9 other models in terms of confusion matrices, precision, accuracy, error rate, recall, f1-score, total trainable, non-trainable parameters, and time consumption to see how well it improved.
xii. Unseen testing data of 130 photo images and 24 real-time video unseen images improved the research's ability to understand. This idea results in advancements in the computer vision area.

In the remaining part of the article, the following is written: Section 2, describes the relevant work. Section 3, discusses the training, testing and presentation of data distribution utilized in this research. Section 4, describes the modified MNV2 design used in this study and suggested technique for masked face classification and identification using the CMNV2 method. Section 5, presents a discussion of how well the suggested approach worked when applied using the confusion matrix and a number of additional performance metrics parameters. The findings of the MFAI and MFGI classifying and accuracy prediction were briefly addressed. Finally, in Section 6, the paper's conclusion is provided.

2. Related work

A deep learning model uses attentional and residual convolutional networks to properly detect faces ages and genders. This system uses attention to pick the most important and detailed face cues for more accurate predictions (Abdolrashidi, Minaei, Azimi, & Minaee, 2020). A novel convolutional neural network-based end-to-end method detects gender and age in real-world events without preparation. CNN architecture involves feature extraction and classification (Agbo-Ajala et al., 2020). A deep convolutional neural network uses a Haar Cascade classifier to extract information from a face to determine its gender, with the limitation of labeled data achieved 89.5% accuracy (Patil, Patil, & Tatkare, 2020).

Image processing is used to analyze information from post-processed facial photos of people of different ages to create a unique model. This process incorporates edge-detection algorithms, binary mask creation and wrinkle density assessment. Using a dataset from the Bangladesh university of engineering and technology (BUET) face database, the Naive Bayes classifier with preset thresholds predicts classes for men and women of all ages with the accuracy 76.3% to 86.6% (Rahman et al., 2020). With strong theoretical guarantees for rank-monotonicity and consistent confidence rankings, the consistent rank logits (CORAL) model reduces prediction error significantly compared to the reference ordinal regression system in numerous face-image datasets. Any advanced DNN classifier on ordinal regression issues benefits from the experimental evaluation rank-consistent technique caused six classes to be incorrectly identified, with a particularly high rate among females of children and elderly women (Cao, Mirjalili, & Raschka, 2020).

The study suggested an externally robust DL-based gender and age identification method. Improved CNN design and learning process, combining GoogLeNet's inception module and a CNN with six (6) hidden layers, increased the algorithm's accuracy (Lim, 2020). In addition, an OpenCV-based model using Keras is developed to categorize humans as male or female facing a camera. The program uses the identification of faces to assess gender and age, allowing customized advertising. This method is used in human–computer interaction (HCI), organization profiling, monitoring and security. The model's limitation is that Anti-surveillance camouflage is confused with facial identification algorithms. The disguised people who were incorrectly identified as either males or women (Mhatre, Morais, & Gurvir Singh, 2020).

Age and gender identification using ResNet50 convolutional neural networks is suggested. The CNN extracts deep features and outputs age and gender features. Evaluation arguments do not affect training or validation and gender identification accuracy is maintained throughout all test data inside the framework. The limitation is after iterations of 20 the validation loss is increased and achieved only 80% of accuracy (Meghana, Sudhakar, Arumugam, Srinivasan, & Prakash, 2020). Anticipated ages are within a factor of five of actual ages. Cloud computing is being used to detect masking conditions. Facial recognition technologies are becoming vital for monitoring and managing access in fields such as law enforcement, medical, education, military services, business and personal assistant applications. Applying the approach to faces in shopping malls, streets, schools and popular sites reveals names, ages, genders and emotions (Yilmazer & Solak, 2020).

A proposed deep learning network, FaceMaskNet-9, can age a person from an image. Additionally, a deep learning-based "age detector" model is used. FaceMaskNet-9 improves accuracy, categorizing masked people by age. The age detector model can limit social media access by age. It also works in surveillance footage, medical diagnosis, advertising, video editing and more achieved only 94.96% of overall accuracy (Golwalkar & Mehendale, 2020).

A CNN-based age and gender prediction method for photo facial images and live video streaming. Three CNN network architectures were tested on internet movie database (IMDB) and web-based collaborative platform (WIKI) datasets, each with a distinct design, filter size and convolution layer size. CNN networks improved identification performance and accuracy. The limitation is it requires lots of time to train the network to correctly predict utilizing the suggested design (Benkaddour, Lahlali, & Trabelsi, 2021). An alternative method combined CNN with an extreme learning machine (ELM) for model training. CNN extracted features from input images and ELM categorized intermediate outputs. The new Audience benchmark for age and gender showed that the new architecture overcomes current methods. The limitation is CNN+ELM lost accuracy, which reached an overall accuracy of 91.4% with the small size of the image setup (Micheala & Shankar, 2021). A Deep Learning model using the Audience dataset was used to produce a gender and age detector that can identify people in images. This technique is used in CCTV surveillance, police work, dating apps and marriage websites (Saxena, Singh, & Singh, 2021).

The gated residual attention network (GRANet) predicts age and gender from facial images using deep learning. It adds a gateway to the residual attention network (RAN) design. Age prediction is a regression challenge, whereas gender identity is a binary classification problem. To improve accuracy, the regression issue can be divided into discrete classification and regression sub-problems. The limitation is when both genders have many similar facial features, identifying a child's gender is more difficult. When images are partially hidden and a person is wearing a hat, glasses, or wig, the model is not able to correctly predict and achieved only 82% of overall accuracy (Garain et al., 2021). Another CNN model uses OpenCV for face recognition and gender classification. Through integrating AI methods, program hardware needs can be reduced. CNN, OpenCV and the Audience dataset consistently perform well throughout training and validation. The limitations are not able to be accessible by a web or mobile application. Cannot identify the gender and age of individuals in a photo, making it simpler for it to recognize massive groups of people (Mahija Kante, Manasa, & Meghana Emandi, 2021).

A multilayer perceptron (MLP) has been suggested for age and gender identification research. The suggested MLP architecture uses skip connections and whitens inputs before linear layers to increase performance. The whitening method addresses dropouts, making it acceptable for Bayesian inference estimates (Kim, 2021). An image package technique designed using OpenCV and Python programming technology and employing the Haar application in real-time (HAAR) cascade has been implemented for real-time face detection using an internet protocol (IP) camera. The limitations are only three features are used: four-rectangular, line, and edge. Additional features required to improve identification and classification (Harshitha, Soujanya, Suhas, Swathi, & Shashikala, 2021).

The DL framework's CAFFE model architecture and DCNN model learning improve age and gender identification accuracy. This reduced neural network design provides an efficient age and gender estimate standard even without learning data (Sai Teja Challa & Ruchitha Reddy Reddy, 2021). CNN is used as the major technique to show the accuracy of classifiers in age and gender identification for feature extraction and learning. Five (5) hidden layers and 5-fold cross-validation improved accuracy. The study also examined how masks affect social interaction comprehension across age, gender and expression (Utkarsha Kumbhar, 2021).

A study discovered that masks affect mood, gender, age and identity. The limitations of the research include multiple viewings of the same photos, making viewers to identify the same people using part-based indications such as mask size relative to the face (Fitousi, Rotschild, Pnini, & Azizi, 2021). Two lightweight CNNs from ImageNet and two mobile-device facial recognition frameworks were used to overcome issues faced by face masks. The research addressed partial face blocking from masks and estimated age and gender from selfies with occluded faces. The limitation is in comparison to gender identification, the accuracy remained poor, with the current database having an overall 93.52% accuracy rate (Alonso-Fernandez, Hernandez-Diaz, Ramis, Perales, & Bigun, 2021). A statistical study of masked people's emotions was done in another way. This was done by generating

the face and gesture recognition network (FGNET)-mask face dataset. A multi-task deep learning model was presented to solve masked face problems. This model uses inputs to determine masked face age, expression and gender (Patel, Nie, Le, & Nguyen, 2021).

After the COVID-19 pandemic, educational institutions have been using computer vision systems for various purposes such as social distance detection, face mask identification, surveillance, crime prevention, emotion recognition and attendance monitoring. Educational computer vision systems involve CNN implementations or other deep learning-based libraries. These devices may benefit both typical schools and emergency scenarios. The subsequent machine vision system uses Microsoft's Cognitive Toolkit deployed on the Microsoft Azure cloud. This technique can detect students' genders, smile abilities, ages, eyes and hair colors (Semerikov et al., 2021).

Linear discriminant analysis (LDA) has been employed to categorize people by gender and age using just facial appearance and a few appearance factors. After identifying gender, the Iterative Dichotomiser3 classifier was used on color images of faces to predict age. Researchers analyzed using the face gesture recognition research network aging (FGR-RNA) dataset. The limitation is using a smaller database overall accuracy is achieved only 90.93% (Shaker & Al-Khalidi, 2022). In another method, a unique algorithm effectively classified people by age and gender. The unique identification method used a CNN architecture to detect age and gender from visual information. The median average accuracy and inference time of machine learning methods used to detect objects were further evaluated (Karahan, Lacinkaya, Erdonmez, Eminagaoglu, & Kasnakoglu, 2022).

One masked facial image can identify many heterogeneity features using deep multi-task learning (MTL). Additionally, a mask-induced variability dataset was developed by quickly superimposing several mask sizes, shapes and textures with an overall accuracy of 89.13% (Mukherjee et al., 2022). Facial masks affected face identification. The study observed a more pronounced decrease in performance in children compared to adults, especially when task complexity was changed. Age and the Cambridge facial memory test-kids (CFMT-K) impacted upright face results in mask and no-mask scenarios. Interestingly, children processed hidden and unmasked faces differently than adults. This study's limitation is that the CFMT-K has more male faces. Higher sex/gender diversity in images of faces resulted in greater gender differences between men and women, resulting in overall 90% accuracy (Stajduhar, Ganel, Avidan, Rosenbaum, & Freud, 2022). Another research examined how young individuals processed masked and unmasked faces in four tasks: emotion identification, popular face recognition with naming, age prediction and gender categorization. Using a face mask lowered performance in identifying well-known people and emotions, but not age or gender (Wong & Estudillo, 2022).

For age identification, a hierarchical regression technique was developed. This method uses a multi-level approach, ranging from group to individual face images, unlike a single regression technique. The technique aims to accurately detect age using feature space across age-specific categories (Islam & Baek, 2022). Additionally, a deep learning-based feature-fusion architecture has been created for the masked face identity proposal. To compensate for data loss from face masks, this architecture combines approximated age, race and gender into the face visualization. The algorithm generates a list of likely identification numbers for a person based on a masked facial image. The limitation is the Performance on the bare face test dataset reduced from 95.52% to 72.54%, a 24.06% decrease (Tipajin Thaipisutikul & Chih-Yang Lin, 2022).

To measure impartiality in actual images, a method has been created and used to modify ones. The research also showed how biassed datasets affect biometric identification system equity. Skewed databases are caused by data-gathering practices that create class imbalance. Adding synthetic images to classes with fewer occurrences balances the dataset for machine learning model training (Lai, Shmerko, & Yanushkevich, 2022). Another technique uses the eyes and forehead to train models to identify masked faces. The research used transfer learning to extract 12 soft biometric traits from the flickr faces HQ (FFHQ) database with 92.5% accuracy and 25 from the celebface characteristics (CelebA) database for masked faces with 87% accuracy. MobileNetV2, AlexNet, SqueezeNet and ResNet50 were designed to do this (Banati, Prakash, Verma, & Srivastava, 2022).

The article discussed two tests on the impact of face masks on age perception. Studies show that smiling faces, even when masked, are seen as older than neutral ones (Ganel & Goodale, 2022). Another investigation examined how masks and sunglasses affect face matching and social judgments of trustworthiness, competence and appearance. Research indicates that masks decrease identification matching in super recognizers, although prosopagnosia individuals wearing masks and sunglasses did not differ from controls. Research indicates that wearing face masks greatly alters perception, highlighting the importance of facial information, regardless of exposure, due to the limitation of reduced dataset achieved overall accuracy 92.4% (Bennetts, Johnson Humphrey, Zielinska, & Bate, 2022). A global tweet dataset was created to identify face masks during the outbreak. The face mask detection (BAFMD) database intends to reduce prejudice by giving additional images from disadvantaged ethnic backgrounds and age groups (Kantarcı, Ofli, Imran, & Ekenel, 2022).

To determine gender in facial images, transfer learning and the EfficientNet framework are used together. Current gender-detection algorithms are typically successful, although they can encounter challenges in certain situations. In the proposed technique, masked facial images from existing datasets are used to train networks, potentially boosting system performance, particularly in face mask situations (Mosayyebi, Seyedarabi, & Afrouzian, 2023). In another research, 3D facial data was utilized to develop a nose clip for a half-face mask. For each participant, a digital nose clip prototype was customized to their unique facial features. Both the half-face mask with and without the nose clip were tested for user satisfaction and air leakage. Users reported improved satisfaction and productivity with the nose clip-enhanced half-face mask (Tjolleng, Chang, Park, & Jung, 2023).

MobileNetV2 and DeepLabV3 were used to create a lightweight image segmentation network. Additionally, the DeepLabV3+ network structure's atrous spatial pyramid pooling (ASPP) module parameters were modified to prevent interference by noise, light, shadow and other factors during segmentation for long-length targets and a multi-feature fusion structure was used instead of the parallel structure in ASPP, allowing the network to obtain richer image features (Tang et al., 2023; Wu et al., 2023).

3. Experiments

This strategy of our study focuses mainly on information presentation, allowing researchers to get more detailed insights from the dataset by seeing them in tabular form, as seen in Table 1. Faces of people without mask belonging to 16 different classes of original dataset samples created by "Sanjaya Subedi" is downloaded from the Kaggle (Subedi, 2018), is shown in Fig. 1. A total of 12,160 artificial masked face images of data that include 16 classes are separated into a training set and a testing set in this research paper, with the first category having 80% of the data and the second 20%. Through data presentation, a researcher gets to know much about the dataset's more in-depth details and distributions utilizing the ability to go deeply into the particulars of each class.

3.1. Training data distribution

The model's predictive capacity for decision-making is learned and improved with the use of training data, as demonstrated in Table 1. With 80% of the data from 9734 images set up for the training of 16 classes, the approach has access to an incredibly large quantity of

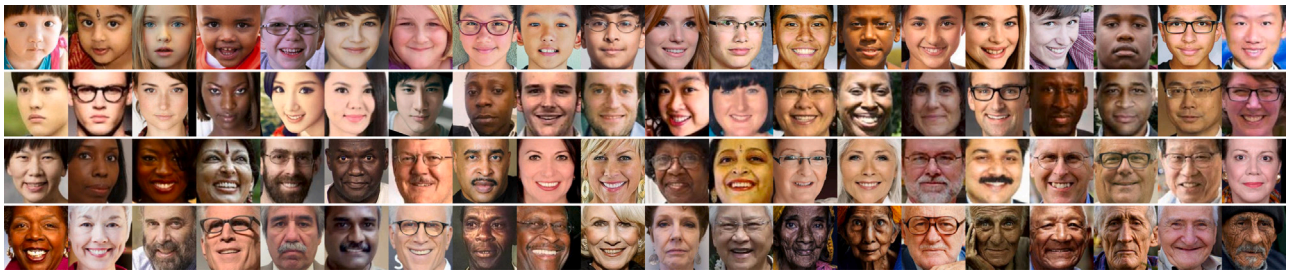


Fig. 1. Original sample images of the dataset without face mask.

Table 1
Classes of 16 different used in this research paper.

Sl.no.	Class number	Name of the class	Training dataset (80%)	Testing dataset (20%)	Total dataset
1	Class1	Female_(0–10)	580	144	724
2	Class2	Male_(0–10)	595	148	743
3	Class3	Female_(11–20)	634	158	792
4	Class4	Male_(11–20)	580	145	725
5	Class5	Female_(21–30)	629	157	786
6	Class6	Male_(21–30)	592	148	740
7	Class7	Female_(31–40)	631	157	788
8	Class8	Male_(31–40)	600	150	750
9	Class9	Female_(41–50)	594	148	742
10	Class10	Male_(41–50)	602	150	752
11	Class11	Female_(51–60)	621	155	776
12	Class12	Male_(51–60)	634	158	792
13	Class13	Female_(61–70)	615	153	768
14	Class14	Male_(61–70)	618	154	772
15	Class15	Female_(71–80)	594	148	742
16	Class16	Male_(71–80)	615	153	768

data, improving its capacity for identifying smaller patterns and relationships. The stronger a model's ability to apply its learning process to new, unknown data, the more data it needs to train on. A bigger training dataset can usually improve the quality and predictability of a model. Through doing so, the model could get a deeper understanding of every feature and pattern in the data, improving its predictive capacity.

3.2. Testing data distribution

As indicated in Table 1, the testing set consists of 20% of the total data of 2426 images utilized to test our model. This stage is critical for determining a model's accuracy and predictability. Through dividing current occurrences from the model's training set, testing data provides an accurate prediction of how well the model will probably work in reality. The testing data is used to identify and evaluate any difficulties, such as overfitting or underfitting, that might have occurred during training in order to verify that the model's predictions are correct and reliable for new, unknown data. Furthermore, depending on its performance on this test dataset, the model's adaptability and efficiency can potentially be fine-tuned and improved. Overall, setting apart 20% for testing data is crucial for validating and developing model, ensuring they are suitable for many applications and real-world decisions.

Our proposed CMNV2 technique was created utilizing a deep neural network architecture, TensorFlow, Keras and OpenCV libraries to achieve MFC, MFAI and MFGI on photo and real-time video images. All of the text (TXT) files, deploy.prototxt, opencv_face_detector.pbtxt, age_deploy.prototxt, gender_deploy.prototxt, along with mask_detector.model, res10_300 × 300_ssd_iter_140000.caffemodel, age_net.caffemodel and gender_net.caffemodel weights are saved in separate files that are included in the proposed model. The CAFFE model uses weights for each layer, in addition, the prototxt and pbtxt files provide the model's layer architecture.

A total of 130 people willingly filled out a Google form to help us test our suggested CMNV2 model on an unseen dataset. The goal of this significant and realistic dataset is to evaluate the model by testing the masked faces of unknown data and individual responses. Through analyzing results from this dataset, we want to analyze the model's predictability and capacity to effectively understand and respond to user input, which will be helpful to overall performance in the real world. This in-depth analysis will provide a further understanding of the model's capability and use under a number of different scenarios. The link created of the Google form https://docs.google.com/forms/d/19YXtbOnhMy0v3EmS6DJ-xCcWSU56BFqZnh7Fx5rATr4/edit?pli=1 and responses of the people who are participated willingly https://docs.google.com/forms/d/19YXtbOnhMy0v3EmS6DJ-xCcWSU56BFqZnh7Fx5rATr4/edit?pli=1#responses. The output predictions of our CMNV2 model on photo images on these unseen dataset is shown in Fig. 8, during the performance evaluation.

4. Proposed methodology

In our research, we designed a modified MNV2 model on top of the standard MNV2 (Howard et al., 2017) structure, adding 8 additional layers while freezing the original MNV2 model all layers in addition to utilizing a transfer learning strategy for feature extraction.

4.1. Modified MobileNetV2

Transfer learning is the process of taking advantage of a model trained on one task is adapted for a second related task using a smaller dataset. In traditional all models are built from scratch and trained on a large number of labeled data to complete a specific task. Transfer learning is the process of using a ML and DL models that has been trained on a particular task over time, can transfer what was learned before. This is particularly useful when there is insufficient labeled data to train a high-quality model.

i. A model of traditional MobileNetV2 and CAFFE is first trained on a huge dataset and the task it will be used for later in pre-training. As this model has previously been trained, these are utilized to extract general features and patterns from the data of our masked face age and gender identification.

Table 2
Added layers of proposed model.

Layer name	Dimension of output	Input number of parameters
Input layer	(224,224,3)	0
MobileNetV2 (frozen layers)	(7,7,1280)	22,70,794
BatchNormalization	(7,7,1280)	8960
GlobalAveragePooling2D	(1280)	0
Dense	(512)	6,55,872
Dense_1	(256)	1,31,328
Dense_2	(256)	65,792
Dense_3	(128)	32,896
Dropout	(0.5)	0
Dense_4	(16)	2,064
Trainable Parameters	**8,90,512**	
Non-trainable Parameters	22,64,384	
Total Parameters	31,54,896	

ii. Using our dataset of 12,160 images as the source domain, the pre-trained model combination of modified MobileNetV2 and CAFFE, such our CMNV2 model is then fine-tuned for required task. Utilizing addressing the specific features of the new task as MFAI and MFGI, the model is able to fine-tune its primary focus and performance. The domain contains a large amount of labeled data and is the primary source for model training. The goal is to apply the information from this source domain to improve the model's performance on a new, differently distributed dataset.
iii. The target domain is the dataset of 130 participants filled Google form to test the model on photo images. Further 24 participants belonging to B.Tech students and Ph.D. scholars volunteered to test our novel CMNV2 technique on real-time video image processing and prediction. In both cases our proposed model performed well, yet it often contains fewer labeled instances. In terms of distribution, this domain differ from the source domain. Our goal is to improve the model from the source domain such that it performs well on the target domain, even if it is not fully represented during training.

Below is an in-depth explanation of the architecture, as well as a simplified Fig. 2, is given. The addition of these additional customized layers in the design, as shown in Table 2, improves the performance of the modified MNV2 framework. The next level of CAFFE (Jia et al., 2014) combines with a modified MNV2 model for MFAI and MFGI. CAFFE is a deep learning structure used for the training and deployment of deep neural networks. It is well-known for its speed and adaptability, making it suited for a wide range of computer vision applications. The improved MobileNetV2 architecture is based on an imported pre-trained MobileNetV2 structure from ImageNet (Li, 2006) of Keras (Chollet, 2015) applications.

An in-depth discussion of the architecturally important aspects of layers of Fig. 2 and Table 2 is provided below:

i. Input (224 × 224 × 3): The input is a 224 × 224 pixel image having three color channels red, green and blue (RGB). This image represents the design's starting point.
ii. Frozen MobileNetV2 Layers: These kinds of layers are part of the MobileNetV2 design which serves as a feature extractor. MobileNetV2 is a DCNN that is optimized for image categorization. These layers are "frozen", indicating that their weights do not change while training. They capture wide visual features such as edges, textures and shapes. All of this beneficial information is used in our work by using the transfer learning technique without training them at all times.
iii. BatchNormalization: This layer normalizes the earlier layer's output. Normalizes the input data, ensuring that the mean activation is close to 0 and the activation standard deviation is close to 1.
(a) Function: Adjusts and scales the activations to normalize the input data. This contributes to reducing the vanishing/exploding gradient issue during training.
(b) Role: Through decreasing internal covariate shift, it stabilizes and speeds the training process. It enables the model's layers to learn more independently, resulting in quicker convergence.
iv. GlobalAveragePooling2D: After the BatchNormalization layer, a GlobalAveragePooling layer was applied. This layer is used for reducing the spatial dimensions of the MobileNetV2 layers feature maps. It takes the average of each feature map and produces a fixed-size feature vector. This stage is important because it ensures a uniform feature presentation irrespective of the size of the input image.
(a) Function: Performs global normalization on the total dataset. It computes the mean and standard deviation for each batch.
(b) Role: Improves the model's capacity to grasp long-term relationships and global patterns in data. It is very useful when working with different datasets with varying features.
v. Dense (512): This layer is constructed from consisting of 512 neurons and is designed to learn complicated features and patterns from a fixed-size vector of features.
(a) Function: Describes a completely linked layer of 512 neurons, with each neuron from the previous layer connected to each neuron in this layer.
(b) Role: As a feature extractor, it learns complicated hierarchical features from incoming data. The model's large number of neurons enables it to grasp complicated relationships in the data.
vi. Dense_1 (256): A second Dense layer with 256 neurons simplifies the data that was acquired.
(a) Function: Another fully connected layer with 256 neurons that extracts additional specific and complex information from the earlier layers high-level generalizations.
(b) Role: Improves the representation of the input data so that succeeding layers are able to differentiate between different classes of data.
vii. Dense_2 (256): The third Dense layer containing 256 neurons refines the feature-developing process.
(a) Function: Like the previous dense layer, this layer continues the feature extraction process with 256 neurons for additional specific and complex information from the earlier layers.
(b) Role: Further improves and organizes the most important information from the data, resulting in a deeper understanding of the underlying patterns.
viii. Dense_3 (128): The feature-enhancing process is continued with a fourth Dense layer of 128 neurons.
(a) Function: Represents a fully connected layer with 128 neurons that extracts more in-depth features that help the model learn from the input data.
(b) Role: Collects high-level abstractions and prepares data for the final stages of classification. This layer provides a more efficient and detailed representation of the input.
ix. Dropout (0.5): Following Dense_3, a Dropout layer that has a dropout rate of 0.5 has been added incorporating some randomness into the model's learning process. Dropout regularization is introduced by randomly setting 50% of the neurons to zero during training.
(a) Function: Overfitting is avoided by encouraging the development of robust features. This dropout rate signifies that during training, 50% of the neurons in this layer are disabled at random during every iteration of training.
(b) Role: As a regularizer, it allows the network to acquire redundant representations and reduces dependence on individual neurons. This improves the model's capacity to generalize to previously unknown data.

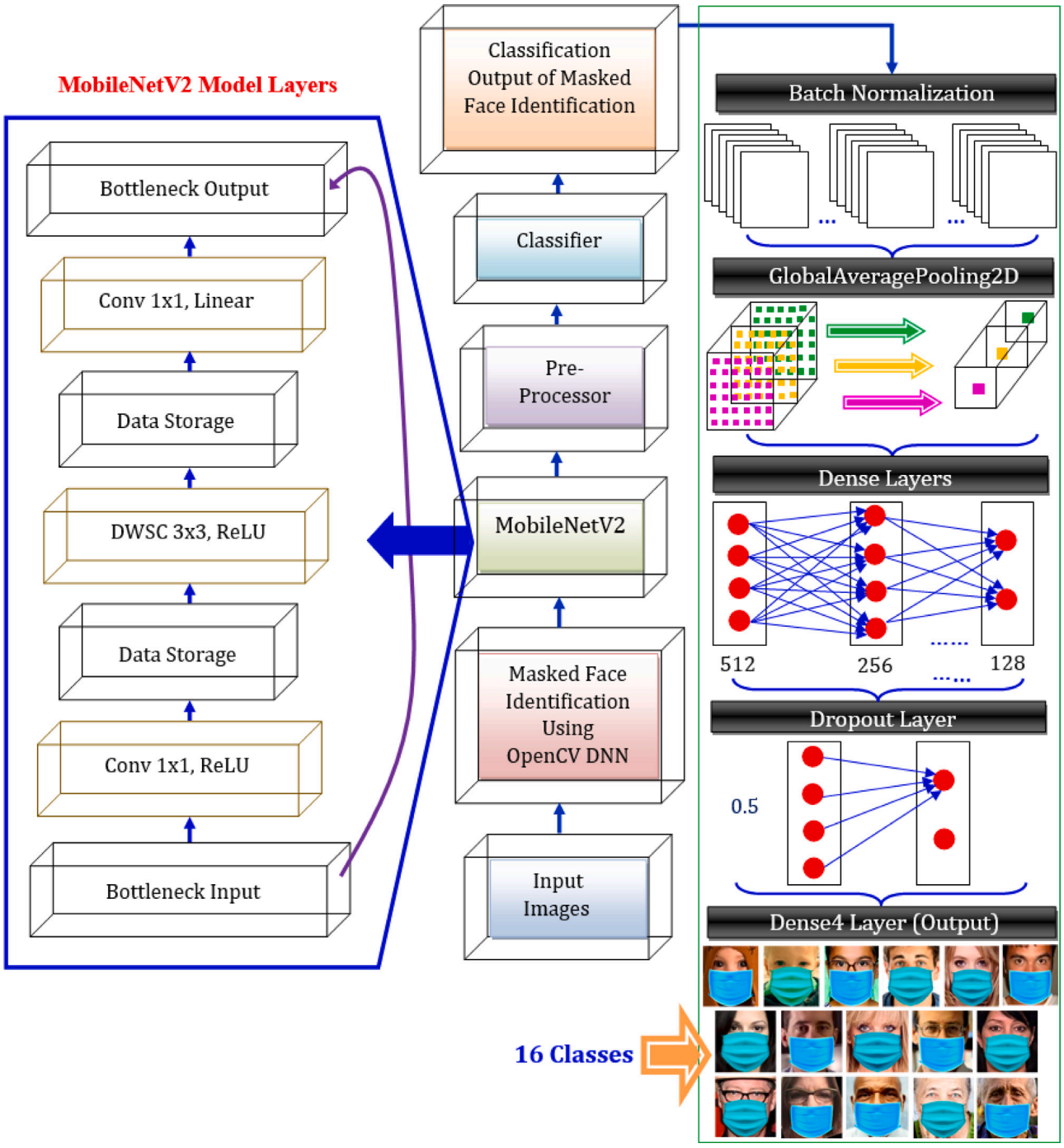


Fig. 2. Modified MobileNetV2 Architecture.

x. Dense_4 (16): The number of neurons in the final, personalized dense layer will be equal to the number of classes to be classified by the model. This layer generates class outcomes for the people's 16 classes, resulting in the model's ultimate output. The model's output is a probability distribution over these 16 classifications. The final dense layer has 16 units, each of which defines how likely the input image is to relate to one of these classes.
(a) Function: Represents the last fully connected layer with 16 neurons, one for each classification task type.
(b) Role: Generates the model's output, which includes the probabilities for each class. Usually, the softmax activation function is used to turn raw outputs into probability distributions, which helps in the categorization of input cases.

The decision to include 8 additional layers, namely BatchNormalization, GlobalAveragePooling, Dense layer 512, Dense_1 layer 256, Dense_2 layer 256, Dense_3 layer 128, Dropout layer (0.5) and a final Dense_4 layer for the output of 16 classes, was motivated by our goal to improve the model's feature extraction and classification capacity for masked face age and gender identification. These layers were selected with maintenance in order to create nonlinearity, prevent overfitting and identify hidden patterns in the data. We went through extensive tests to examine the performance of the modified MNV2 model with different numbers of layers to give novel insights into this design decision. Our findings consistently showed that including these particular layers increased accuracy, convergence speed and adaptability in dealing with the complexity of facial features in our dataset.

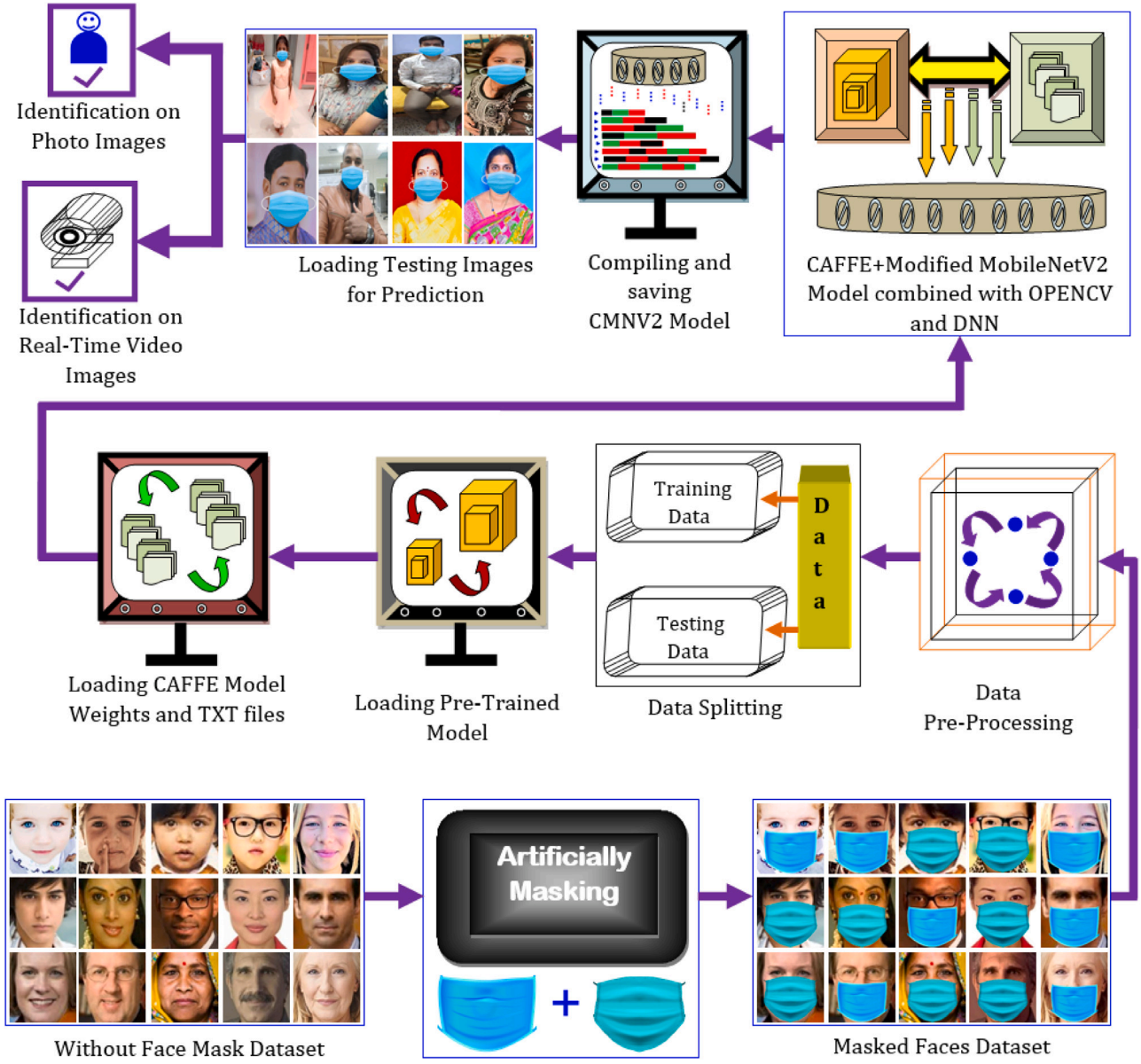


Fig. 3. Block Diagram of the Proposed CMNV2 Model.

The suggested technique utilizes the CMNV2 model (Kumar & Bansal, 2023), for masked face classification and masked face age and gender identification on both photo and real-time video images. Although numerous characteristics can exist in a single image, their boundaries may not seem to be straight. The neural network was given non-linearity after each convolution layer by using an activation function known as a rectified linear unit (ReLU). When the output of the convolution layer is positive, the ReLU function is linear; when it is negative, it is zero. As shown in Fig. 3, of the block diagram, the proposed model was trained and thereafter used for the classification and identification of masked faces of photo and real-time video images.

All of the stages that follow describe the block diagram of the proposed CMNV2 method is explained below.

Phase 1: This study made use of a database of masked facial images, by artificially masking them.

Phase 2: Multi-class relates to 16 different classes, each having images of people with their faces masked. Following that, 16 different folders were established based on their classifications. After the data was pre-processed, a graphical application called "labeling" was used to label the images. Label information is provided in both the training and testing datasets for each image, as detailed in Sections 3.1 and 3.2.

Phase 3: Used stratification to determine the ratio of 16 classes of masked face data in the training and test sets, with a total of 12,160 images.

Phase 4: This previously present phase of images was separated into two different categories as mentioned in Sections 3.1 and 3.2.

Phase 5: After processing the training and testing data, various batches were used to determine the amount of data to analyze before updating the model.

Phase 6: After applying model training to the input data, the methodology was executed and the processed outputs were compared to the test output, as indicated in Section 4.6.

Phase 7: Loaded the modified MobileNetV2 and CAFFE frameworks, then saved and utilized the proposed CMNV2 model on the testing dataset, as described in Section 4.7.

Phase 8: Displaying both photo and real-time video images to evaluate the model's predictions, as explained in detail in Section 4.8.

4.2. Artificially masked face dataset

In our suggested research paper, a method such as artificial masking was used. The original source images of the dataset was created by "Sanjaya Subedi" and is accessible on the Kaggle (Subedi, 2018), which includes people aged '0–116' years belonging to a male and female gender without wearing a face mask. The research in this paper provides an age range of '0–80' years of both male and female categories and created 16 classes, as described in Sections 3.1 and 3.2. The original 12,160 images were downloaded and then artificially masked people's faces using light blue and dark blue color masks, which were then utilized in our study work for training and performance evaluation. The complete 12,160 artificially masked faces of images of the dataset created and used in our research is now made publicly accessible on the Kaggle platform, the link https://www.kaggle.com/datasets/banilkumar20phd7071/masked-face-age-and-gender-identify-artificial-masking is provided here.

4.3. Input data

After downloading, the input the procedure began with a collection of an original dataset containing images of people without face masks. According to our description in Section 4.2, this data obtained provides the foundation of our study. Technologically modern artificial approaches are applied to fulfill the fundamental requirements for data with masked faces. The faces in the downloaded dataset are artificially masked using these techniques, utilizing a light and dark blue color two different mask types and variations. Section 3, describes in-depth how the dataset is distributed. As a result, a large number of images with masked faces have been created, having an extensive variety of reactions to facial emotions, lighting conditions and background colors.

4.4. Pre-processing of data

After successfully masking the faces. The data was saved in a folder for easy access in the next stage. The "pre-processing" function reads every file in the provided directories and scales images correctly to prepare them for usage with the proposed model. Each of the images is preprocessed by adding data once they have been loaded and turned into an array. The images were converted into a Numerical Python (NumPy) library, which allowed the values to be handled quickly throughout the process. The image data was then stored in the label binarizer (LB) to determine the 0 and 1 integer categories to begin analyzing the pattern. The images were then separated into sets for training and for testing.

4.5. Data splitting

As explained in Section 3, the training set of data is used for teaching the model, helping it to learn the correlations and patterns in the data, whereas the testing set is used for evaluating the model's performance and generalized capability. This 80% to 20% split ensures the model's stability and performance in the application by balancing giving sufficient information for training with an in-depth evaluation.

4.6. Training the model

The suggested model design process is shown in Fig. 2. The pre-trained MobileNetV2 model was imported from the ImageNet and Keras applications. Following the loading of the basic model, 8 separate layers are added utilizing a transfer learning technique, as indicated in Table 2 and described more deeply in Section 4. Following this, a new set of trainable CAFFE layers was incorporated to find features that can be used to classify and identify a masked faces. The structure of the model was modified once the weights were stored after loading the CAFFE, text "TXT" and "weight" files. Applying a pre-trained model can let it utilize bias weights with minimum additional computational cost and without losing any previously learnt features. The combination of the modified MobileNetV2+ CAFFE (CMNV2) model, was further trained for 72 epochs along with 64 batch sizes of individual training batches at a learning rate of 0.001. In computer vision and DL applications, the Adam optimizer is used to alter the rate of improvement toward a goal that is desired for each neural network weights. The Adam optimizer estimates the first and second gradient moments. Transfer learning is an excellent way to rebalance neural network weights after each batch. To train and update the DNN model for prediction, photo and real-time video images were utilized.

4.7. CMNV2 proposed model

The CMNV2 model, which combines CAFFE with modified MobileNetV2, is proposed for MFC, MFAI and MFGI. After training, the CAFFE model for FaceNet and the modified MobileNetV2 model for classification are used, in addition, the CMNV2 model is used for MFC, MFAI and MFGI. A blob is a data structure used in DL frameworks such as CAFFE and OpenCV for pre-processing and image transformation. The images are saved in a format appropriate for DNN, which is a multidimensional array. Input data must be standardized before being passed into a neural network and a blob helps to do this. OpenCV is a robust computer vision toolkit that includes a variety of programs and techniques for photo and real-time video analysis. It is used to process photos and videos while working with deep-learning models.

Face identification and recognition models pre-trained by OpenCV can potentially be used to find faces in photos and video frames. OpenCV assists in the extraction of blobs (regions of interest) from images or frames including masked faces. In advance of sending images into the neural network, OpenCV is applied to pre-process them, such as scaling, normalization and color improvements. Following the storage of the frame's dimensions and its creation of a blob function, the deep neural network was trained to identify and predict using the suggested method for masked face age and gender identification. The blob function's input and setup include a list of faces, their locations and a group of predictions.

4.8. Testing the model

Our suggested CMNV2 methodology was loaded in order to identify the masked faces of 16 different classes and predict their facial features and then testing data of photo and real-time video images were uploaded. If the probability of identification on photos is more than 0.5, the masked face will be identified on photo and real-time video images by calculating the X and Y coordinates of the objects bounding boxes. To ensure that the bounding boxes fit inside the photos of face boundaries, class labels of age and gender in green color are presented when the prediction is correct. These were used to produce the text and the border. Combining prototxt files, weights, the blob function and the RGB dimensions of both photo and real-time video image files enabled masked face age and gender identification, which was achieved with the help of a label and bounding boxes. Finally, Figs. 8 and 9, demonstrate the output results for both predicted photo and real-time video mages.

The following important steps were engaged in the testing process, metrics, and evaluation results:

i Dataset Creation: We selected a dataset containing a diverse variety of images of photos and real-time video, ensuring variation in backgrounds, and lighting situations. This dataset is the result of real-world circumstances, and it is used to evaluate the model's robustness.

ii Testing Procedure: The model was tested using both photo and live video environments. Still, images were uploaded into the model for photo-based testing. Real-time image identification in changing conditions was used during live video testing.

Table 3
Performance evaluation of total (T) testing data correctly predicted T_{+ve} and F_{+ve} values of 10 different models.

Class no.	Class name	MobileNet			MobileNetV2			ResNet101			DenseNet169			DenseNet201		
		T	T_{+ve}	F_{+ve}	T	T_{+ve}	F_{+ve}	T	T_{+ve}	F_{+ve}	T	T_{+ve}	F_{+ve}	T	T_{+ve}	F_{+ve}
Class1	Female_(0–10)	144	144	0	144	144	0	144	144	0	144	144	0	144	144	0
Class2	Male_(0–10)	148	110	38	148	147	1	148	134	14	148	135	13	148	147	1
Class3	Female_(11–20)	158	7	151	158	100	58	158	44	114	158	109	49	158	130	28
Class4	Male_(11–20)	145	139	6	145	143	2	145	142	3	145	145	0	145	133	12
Class5	Female_(21–30)	157	153	4	157	155	2	157	109	48	157	120	37	157	141	16
Class6	Male_(21–30)	148	127	21	148	148	0	148	131	17	148	148	0	148	125	23
Class7	Female_(31–40)	157	135	22	157	114	43	157	127	30	157	157	0	157	139	18
Class8	Male_(31–40)	150	111	39	150	85	65	150	120	30	150	138	12	150	87	63
Class9	Female_(41–50)	148	115	33	148	80	68	148	147	1	148	141	7	148	148	0
Class10	Male_(41–50)	150	129	21	150	147	3	150	147	3	150	120	30	150	115	35
Class11	Female_(51–60)	155	114	41	155	101	54	155	110	45	155	84	71	155	151	4
Class12	Male_(51–60)	158	155	3	158	152	6	158	143	15	158	131	27	158	141	17
Class13	Female_(61–70)	153	153	0	153	153	0	153	145	8	153	153	0	153	153	0
Class14	Male_(61–70)	154	145	9	154	136	18	154	145	9	154	153	1	154	146	8
Class15	Female_(71–80)	148	144	4	148	121	27	148	147	1	148	141	7	148	138	10
Class16	Male_(71–80)	153	147	6	153	149	4	153	153	0	153	152	1	153	146	7
Class No.	Class Name	EfficientNetB0			ResNet152V2			InceptionV3			Xception			CMNV2 (Proposed)		
		T	T_{+ve}	F_{+ve}	T	T_{+ve}	F_{+ve}	T	T_{+ve}	F_{+ve}	T	T_{+ve}	F_{+ve}	T	T_{+ve}	F_{+ve}
Class1	Female_(0–10)	144	144	0	144	143	1	144	144	0	144	144	0	144	144	0
Class2	Male_(0–10)	148	146	2	148	147	1	148	144	4	148	133	15	148	144	4
Class3	Female_(11–20)	158	158	0	158	157	1	158	158	0	158	157	1	158	158	0
Class4	Male_(11–20)	145	143	2	145	141	4	145	145	0	145	142	3	145	142	3
Class5	Female_(21–30)	157	141	16	157	156	1	157	153	4	157	154	3	157	146	11
Class6	Male_(21–30)	148	143	5	148	148	0	148	144	4	148	133	15	148	148	0
Class7	Female_(31–40)	157	139	18	157	156	1	157	149	8	157	155	2	157	152	5
Class8	Male_(31–40)	150	133	17	150	84	66	150	135	15	150	136	14	150	118	32
Class9	Female_(41–50)	148	120	28	148	126	22	148	127	21	148	124	24	148	144	4
Class10	Male_(41–50)	150	131	19	150	147	3	150	130	20	150	145	5	150	150	0
Class11	Female_(51–60)	155	141	14	155	99	56	155	129	26	155	116	39	155	136	19
Class12	Male_(51–60)	158	148	10	158	154	4	158	153	5	158	158	0	158	157	1
Class13	Female_(61–70)	153	153	0	153	153	0	153	153	0	153	151	2	153	153	0
Class14	Male_(61–70)	154	92	62	154	151	3	154	151	3	154	139	15	154	150	4
Class15	Female_(71–80)	148	140	8	148	136	12	148	114	34	148	142	6	148	147	1
Class16	Male_(71–80)	153	153	0	153	152	1	153	125	28	153	153	0	153	153	0

iii Evaluation Metrics: To measure the model's performance, we used common evaluation metrics like accuracy, error rate, recall, f1-score, parameters of total, trained, non-trained and computation time. These metrics are crucial for measuring the model's reliability, comprehensiveness, and overall effectiveness in identifying images.

iv Analysis of Confusion Matrix: We created confusion matrices to get information about the model's capacity to accurately categorize and indicate possible areas for development. This study helped in understanding specific cases where the model can have difficulties.

v Comparative Analysis: To assess our model's performance, we compared it to current state-of-the-art methodologies and baseline models. This comparative study sets the stage for evaluating the model's appropriate strengths and problems.

vi Qualitative Results: We also provide visual examples of the model's outputs on real photos and video frames. This qualitative study supports the quantitative measurements, providing a deeper understanding of the model's capabilities.

vii Outcomes: Our model performed well across every circumstance evaluated. The confusion matrices as shown in Table 3, give further insight into the model's face-specific challenges in image identification of each class. Table 4, contains precise data about the quantitative outcomes, such as precision, recall, and f1-score values.

5. Results and discussion

The dataset has 16 classes with a total of 12,160 masked face images, as described in Section 3. Our CMNV2 method on testing data identified masked faces with 96.54% accuracy.

5.1. Setup

This study's experiments were carried out using a Jupyter Notebook. The Jupyter Notebook is a suitable tool for deep learning model training. Experiments in our research were based on the Python 3.9 libraries pip, pandas and scikit-learn, which were all installed. Pandas, a powerful data processing application, imports and then it updates this dataset. Scikit-learn is an ML and DL Model open-source library data analysis that was used to assess the effectiveness of this methodology. This investigation was carried out on a system with 64 GB of RAM, Intel(R) and Xeon(R) W-2125 CPU operating at 4.00 GHz.

A confusion matrix (Room, 2019), is a valuable tool for measuring the successful performance of a classification model in machine learning, deep learning and statistical analysis. The matrix describes a dataset's predicted and actual classes, with four potential outcomes: true positive (T_{+ve}), true negative (T_{-ve}), false positive (F_{+ve}) and false negative (F_{-ve}). T_{+ve} represents the number of times the model successfully predicted a positive class. T_{-ve} indicates the number of times the model successfully predicted a negative class. F_{+ve} represents the number of times the model predicted a positive class wrongly. F_{-ve} indicates the number of times the model predicted a negative class wrongly. The cells in the matrix show the number of correctly or incorrectly identified cases for each class.

5.2. Accuracy

It is a measure of a classification model's overall accuracy. It denotes the percentage of cases successfully identified by the model out of the total number of occurrences. Eq. (1), calculates the classification's accuracy.

$$Accuracy = \frac{T_{+ve} + T_{-ve}}{T_{+ve} + F_{+ve} + F_{-ve} + T_{-ve}} = 96.54\% \quad (1)$$

Table 4
Precision (P)%, Recall (R)%, F1-score (F1)% performance analysis comparison of 10 different models.

Class	Class name	MobileNet			MobileNetV2			ResNet101			DenseNet169			DenseNet201		
		P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Class1	Female_(0–10)	49	100	66	69	100	82	56	100	72	75	100	85	49	100	66
Class2	Male_(0–10)	99	74	85	75	99	86	98	91	94	96	91	93	96	99	98
Class3	Female_(11–20)	100	4	8	100	63	78	98	28	43	100	69	82	100	82	90
Class4	Male_(11–20)	74	96	84	81	99	89	82	98	89	83	100	91	99	92	95
Class5	Female_(21–30)	81	97	88	69	99	81	96	69	80	94	76	84	94	90	92
Class6	Male_(21–30)	94	86	90	77	100	87	98	89	93	94	100	97	95	84	89
Class7	Female_(31–40)	80	86	83	82	73	77	84	81	82	73	100	84	97	89	92
Class8	Male_(31–40)	85	74	79	99	57	72	92	80	86	96	92	94	98	58	73
Class9	Female_(41–50)	86	78	82	99	54	70	71	99	83	77	95	85	89	100	94
Class10	Male_(41–50)	74	86	80	82	98	89	80	98	88	97	80	88	97	77	86
Class11	Female_(51–60)	94	74	83	81	65	72	92	71	80	92	54	68	80	97	88
Class12	Male_(51–60)	99	98	99	99	96	97	99	91	95	99	83	90	100	89	94
Class13	Female_(61–70)	99	100	100	100	100	100	98	100	99	100	100	100	100	100	100
Class14	Male_(61–70)	86	94	90	99	88	93	97	94	95	83	99	90	99	95	97
Class15	Female_(71–80)	96	97	97	100	82	90	91	99	95	100	95	98	100	93	97
Class16	Male_(71–80)	99	96	97	99	97	98	94	100	97	97	99	98	100	95	98
Class	Class name	EfficientNetB0			ResNet152V2			InceptionV3			Xception			CMNV2 (Proposed)		
		P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Class1	Female_(0–10)	100	100	100	100	99	100	96	100	98	94	100	97	100	100	100
Class2	Male_(0–10)	96	99	97	99	99	99	99	97	98	100	90	95	99	97	98
Class3	Female_(11–20)	100	100	100	100	99	100	100	100	100	100	99	100	100	100	100
Class4	Male_(11–20)	90	99	94	99	97	98	93	100	96	88	98	98	97	98	98
Class5	Female_(21–30)	91	90	90	73	99	84	85	97	91	80	98	88	87	93	90
Class6	Male_(21–30)	88	97	92	84	100	91	92	97	94	100	90	95	97	100	98
Class7	Female_(31–40)	87	89	88	87	99	93	88	95	91	87	99	92	96	97	97
Class8	Male_(31–40)	88	89	88	100	56	72	90	90	90	91	91	91	99	79	88
Class9	Female_(41–50)	94	81	87	100	85	92	100	86	92	99	84	91	97	97	97
Class10	Male_(41–50)	89	87	88	78	98	87	95	87	91	90	97	93	84	100	91
Class11	Female_(51–60)	82	91	87	88	64	74	76	83	80	95	75	84	93	88	90
Class12	Male_(51–60)	99	94	96	99	97	98	99	97	98	98	100	99	100	99	100
Class13	Female_(61–70)	99	100	100	98	100	99	100	100	100	100	99	99	100	100	100
Class14	Male_(61–70)	100	60	75	99	98	98	84	98	90	100	90	95	100	97	99
Class15	Female_(71–80)	99	95	97	100	92	96	100	77	87	100	96	98	100	99	100
Class16	Male_(71–80)	77	100	87	100	99	100	100	82	90	93	100	96	99	100	100

5.3. Error rate

This can be defined as the proportion of the number of total misclassifications to the total predictions produced by the model. It is provided with Eq. (2).

$$Errorrate = \frac{F_{+ve} + F_{-ve}}{T_{+ve} + F_{+ve} + F_{-ve} + T_{-ve}} = 3.46\% \quad (2)$$

The confusion matrix represents the performance of a 16-class multiclassification model are examined. The actual class operates in rows, whereas the expected class operates in columns. Actual-predicted class combination occurrence counts are provided as numbers by matrix cells. Off-diagonal rows and columns are incorrect predictions, while diagonal rows and columns are correct. This demonstrates which categories are confusing and where the classification method can be improved. The confusion matrix shows exactly how well the model performs for each class and identifies any patterns of uncertainty. These numbers, which are explicitly presented in the cells of the matrix from Fig. 4(a) to Fig. 4(j), indicate the number of instances that are rightly or wrongly classified for each class.

5.4. Performance comparison of 10 different models

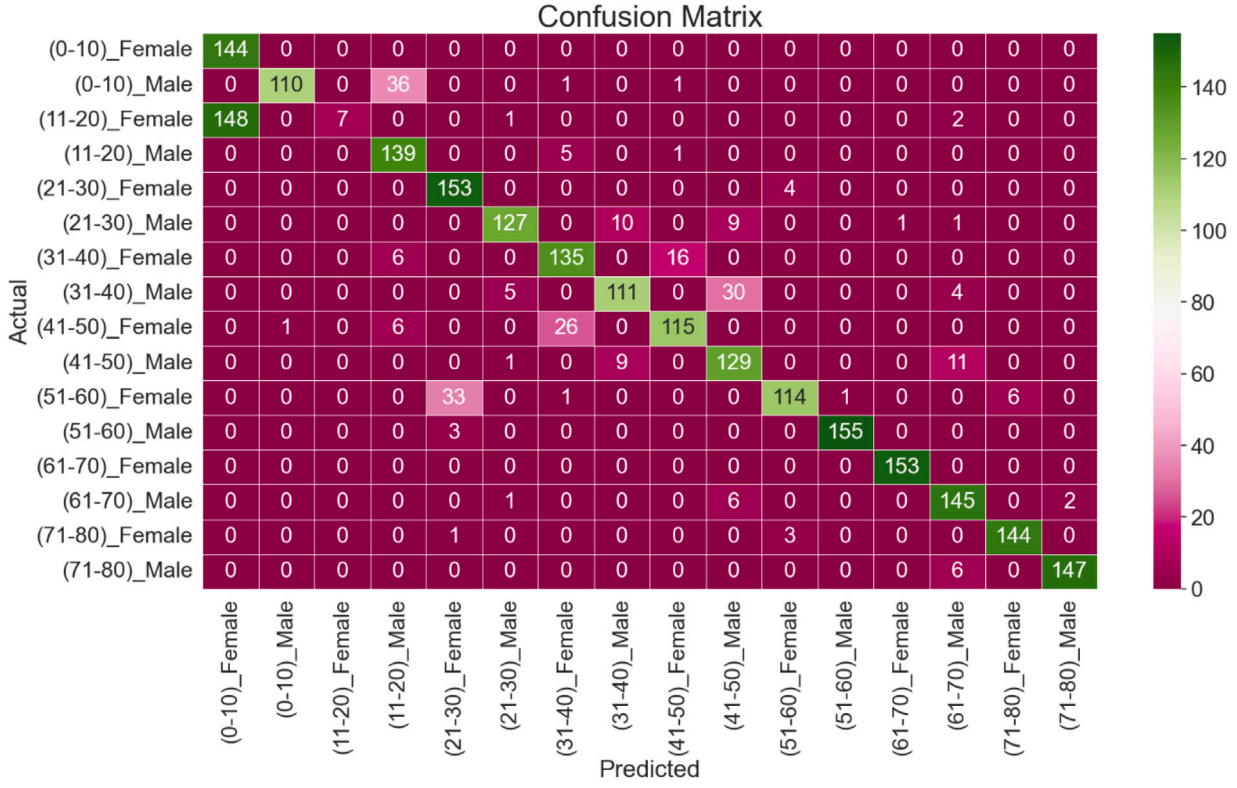
To provide a thorough and organized representation comparison of 10 different models of numerical-related categorical data, all tables are used to represent the obtained data points. Numerical numbers and different information displayed in a tabular form are referred to as the actual data values. On the other hand, all the figures are used to visually represent correlations, trends, and patterns in data across a comparison of 10 different models. A data visualization's characteristics of the representation in the figure relate to the visual elements and design features. To fully understand the data contained in our study paper, both factors are essential.

The results of the 16 classes performance evaluation on testing data related to 10 different models of MobileNet, MobileNetV2, ResNet101, DenseNet169, DenseNet201, EfficientNetB0, ResNet152V2, InceptionV3, Xception and proposed CMNV2 is shown in Fig. 4 and Table 3. Also, performance analysis of precision, recall and f1-score is shown in Table 4 and Fig. 5.

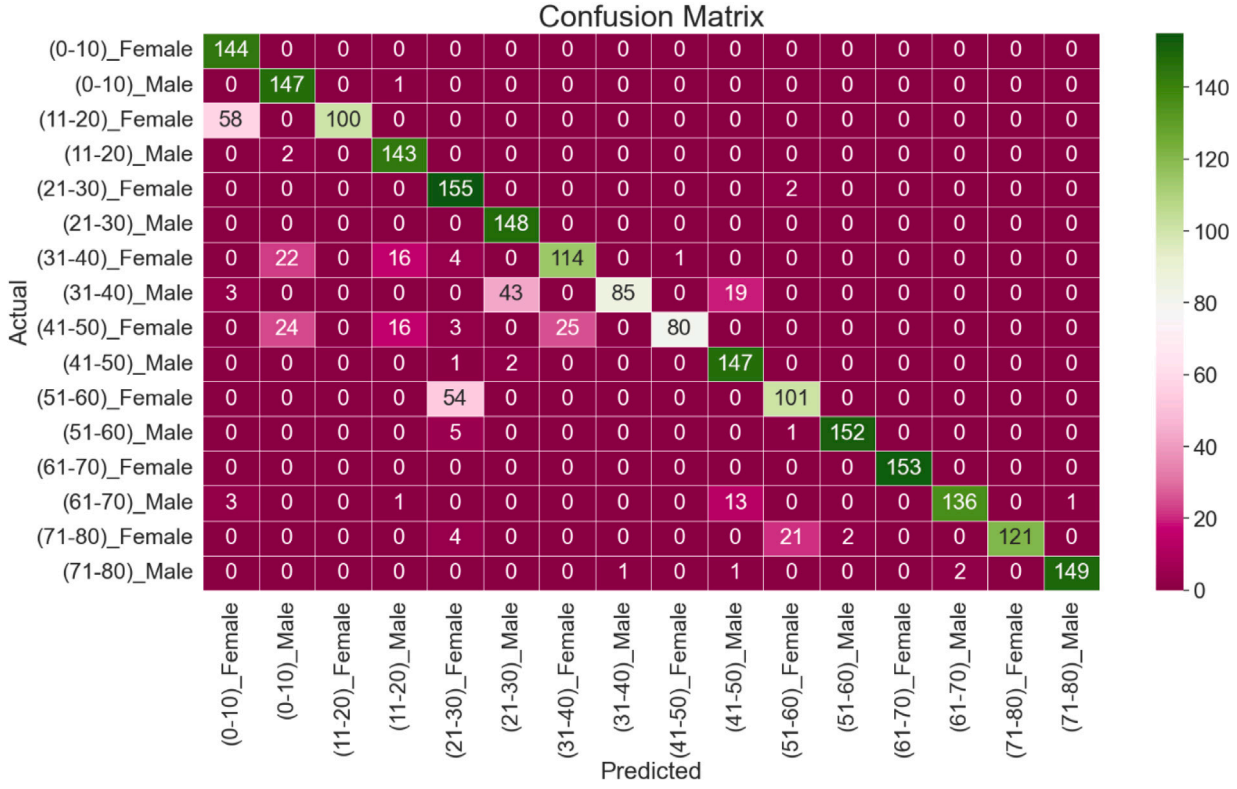
Architectural and operational efficiency of CMNV2 outperforms the other 9 standard models as follows:

CMNV2 model, the total layers are 165, trainable layers are 8, and non-trainable layers are 157, with more efficiency and high performance of 96.54% accuracy. MobileNet model, the total layers are 89, trainable layers are 89, and non-trainable layers are 0, but with a performance efficiency of 83.59% accuracy. MobileNetV2 model, the total layers are 157, trainable layers are 157, and non-trainable layers are 0, but with a performance efficiency of 85.53% accuracy. ResNet101 model, the total layers are 348, trainable layers are 348, and non-trainable layers are 0, but with a performance efficiency of 86.40% accuracy.

DenseNet169 model, the total layers are 598, trainable layers are 598, and non-trainable layers are 0, but with a performance efficiency of 89.49% accuracy. DenseNet201 model, the total layers are 710, trainable layers are 710, and non-trainable layers are 0, but with a performance efficiency of 90.02% accuracy. EfficientNetB0 model, the total layers are 241, trainable layers are 241, and non-trainable layers are 0, but with a performance efficiency of 91.71% accuracy. ResNet152V2 model, the total layers are 567, trainable layers are 567, and non-trainable layers are 0, but with a performance efficiency of 92.75% accuracy. InceptionV3 model, the total layers are 314, trainable layers are 314, and non-trainable layers are 0, but with a performance

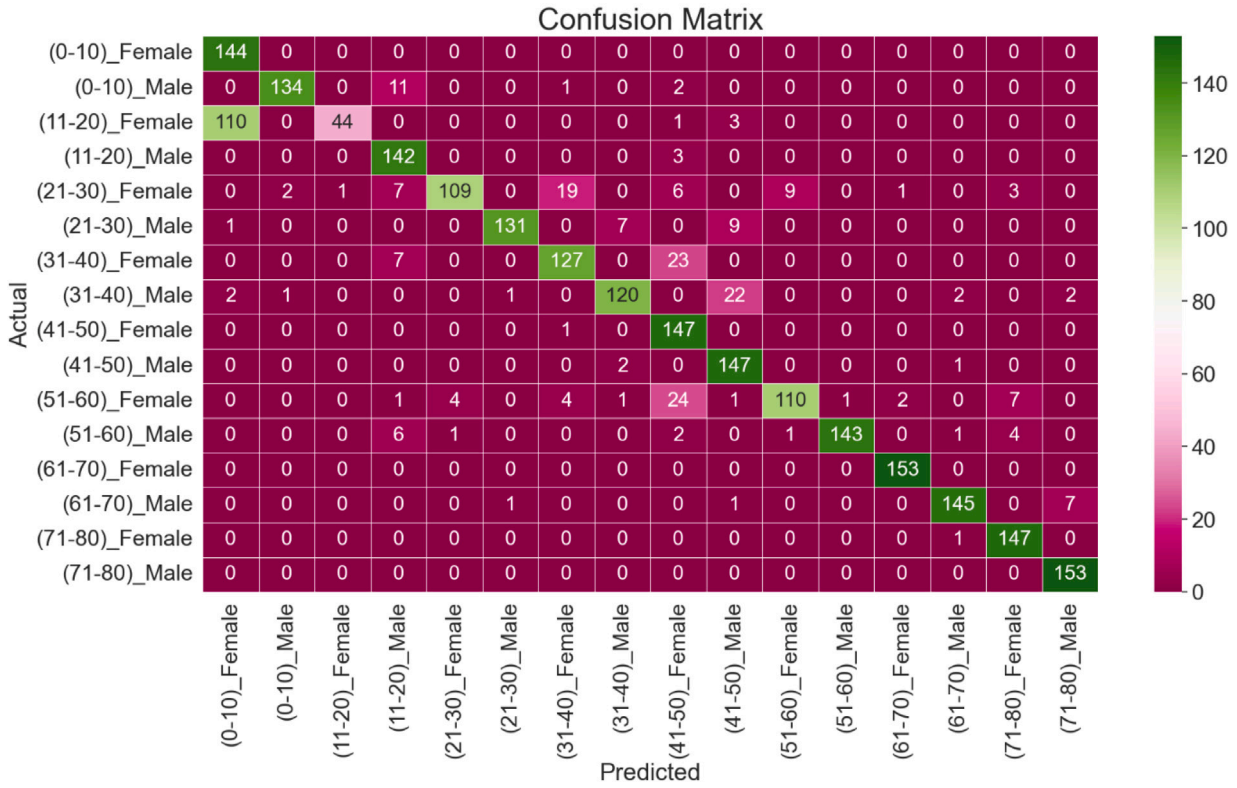


(a) MobileNet

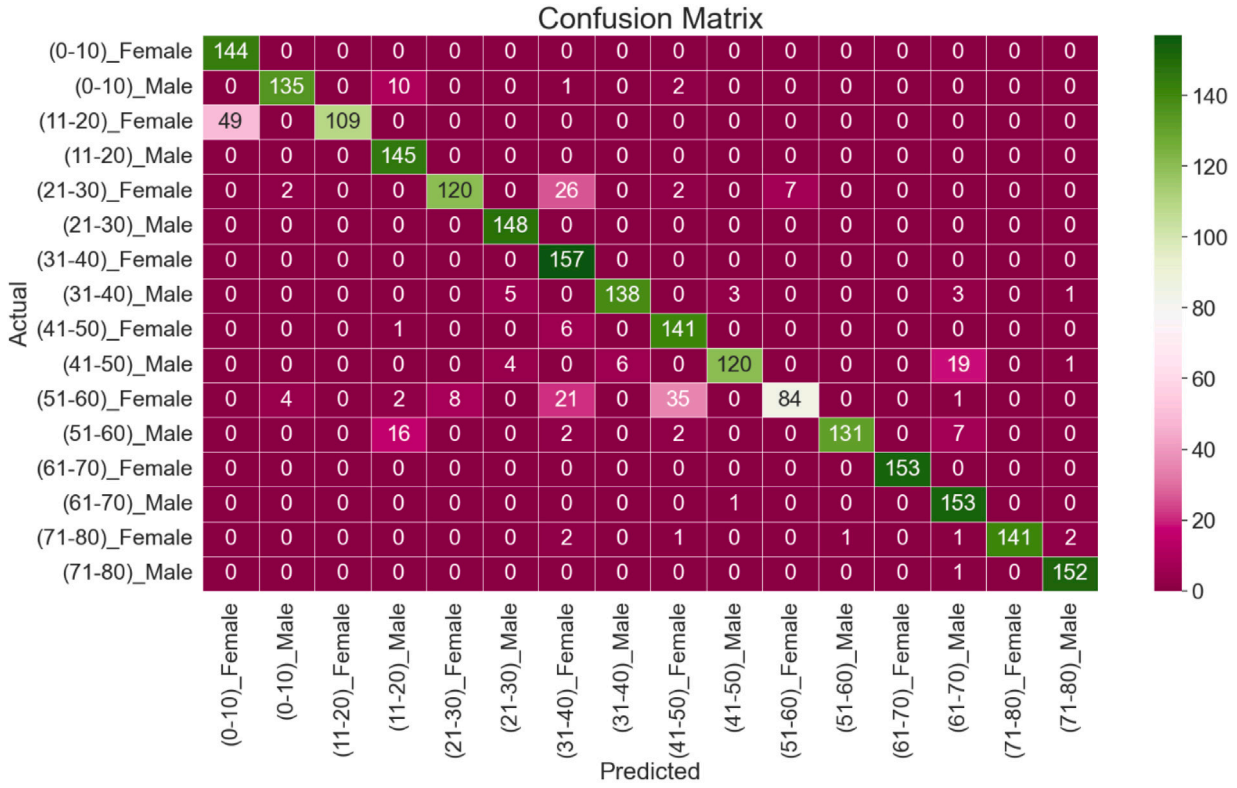


(b) MobileNetV2

Fig. 4. Confusion matrix of 10 different models.

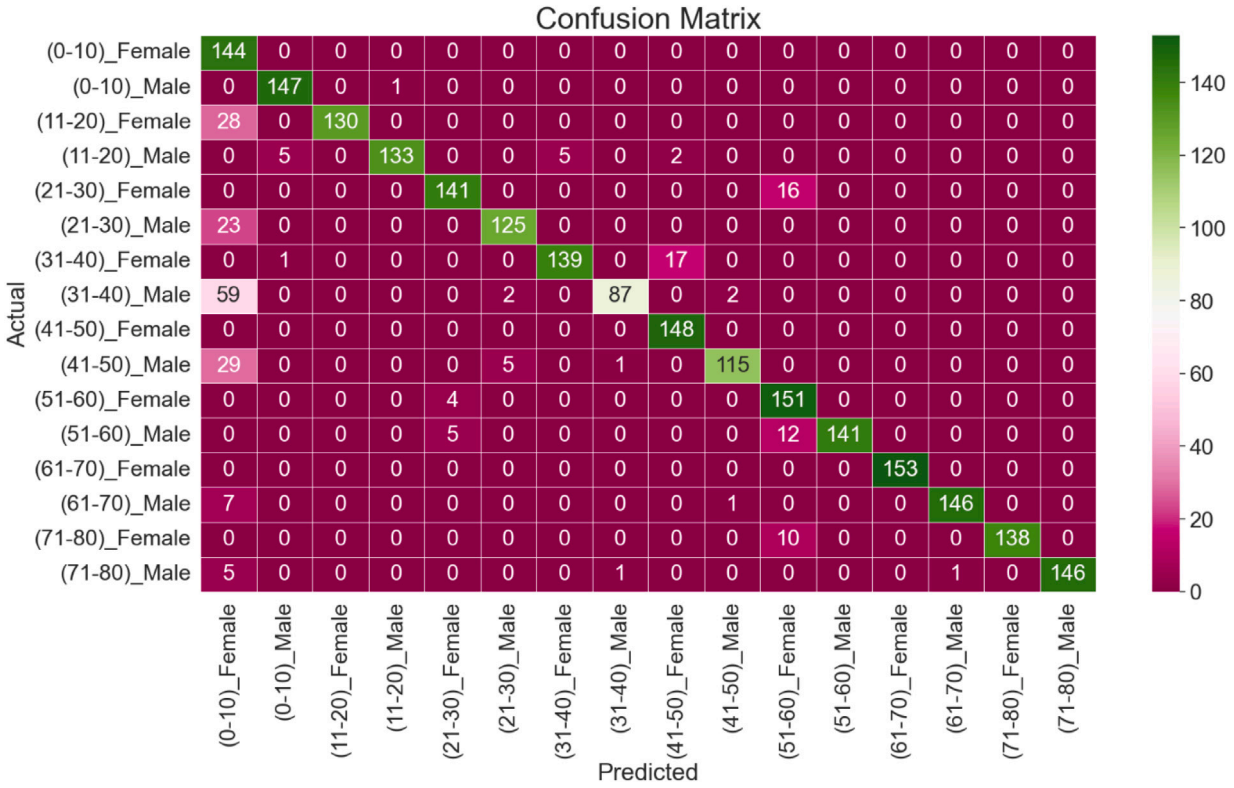


(c) ResNet101

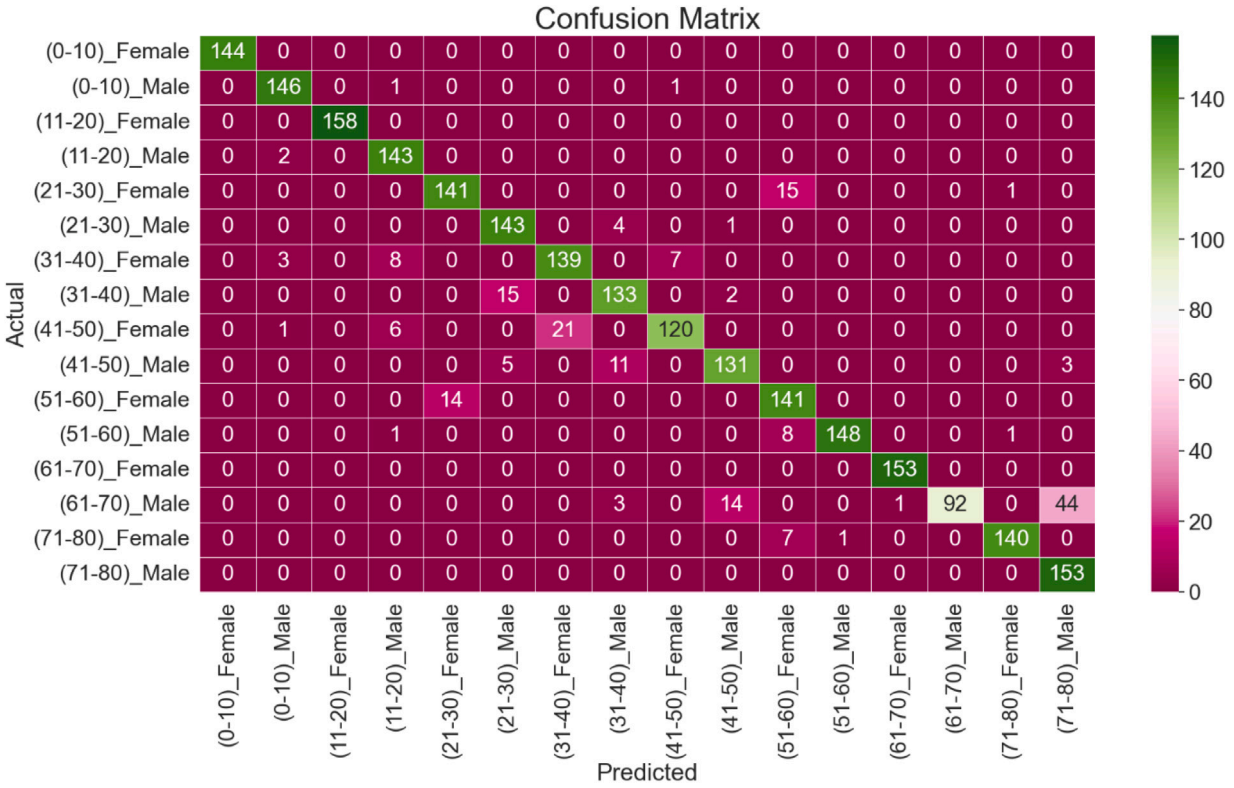


(d) DenseNet169

Fig. 4. (*continued*).

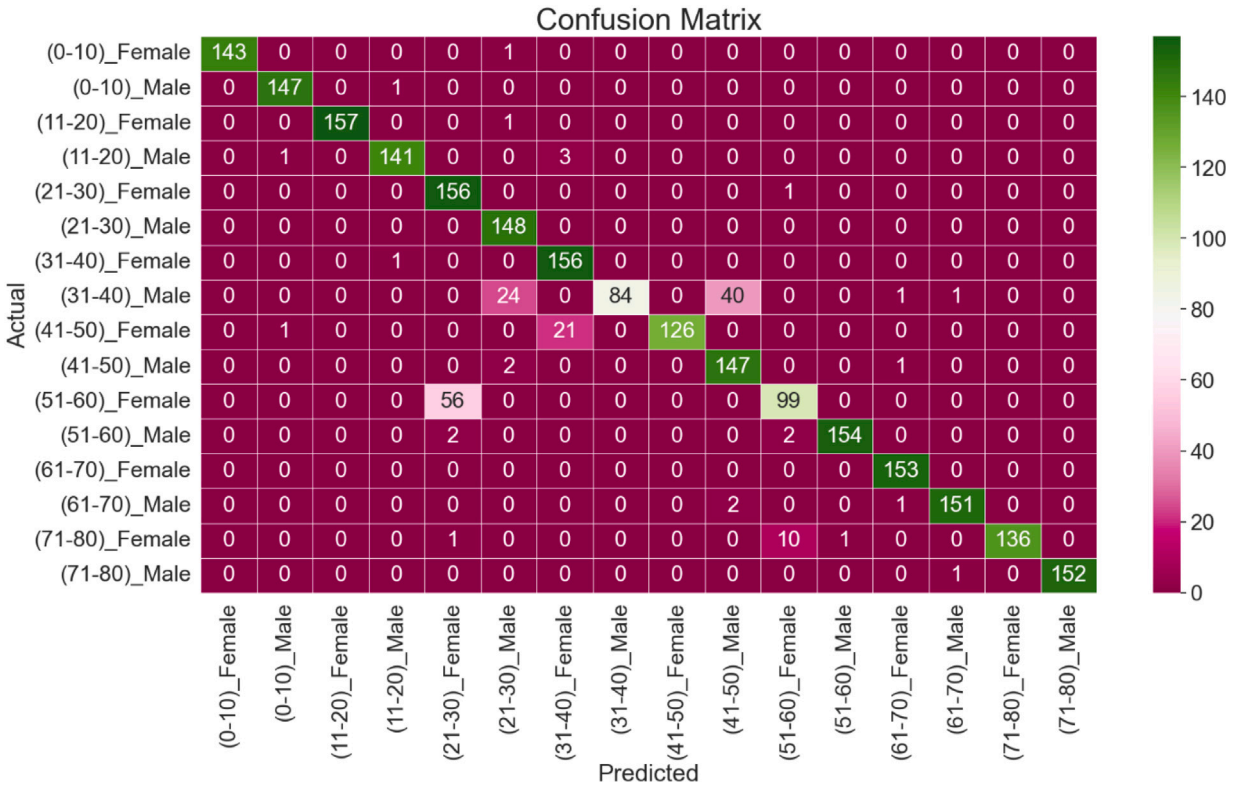


(e) DenseNet201

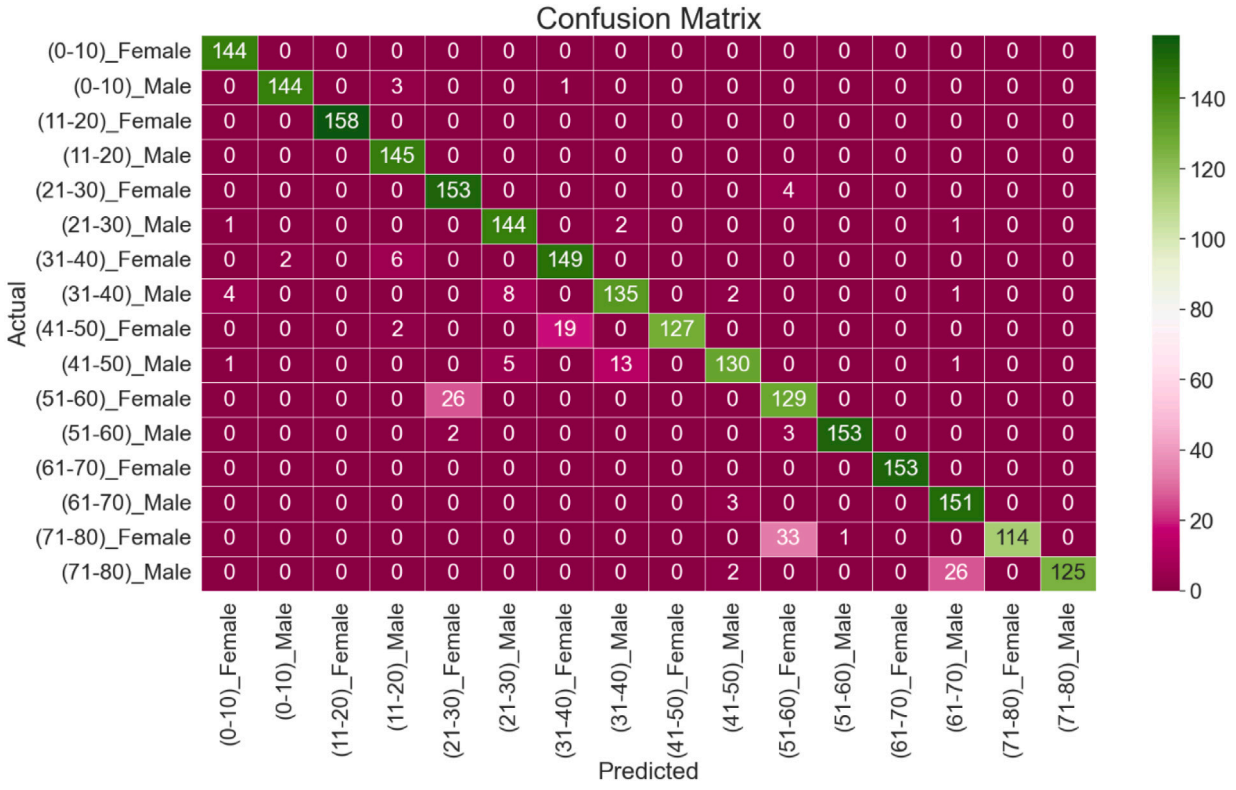


(f) EfficientNetB0

Fig. 4. (*continued*).

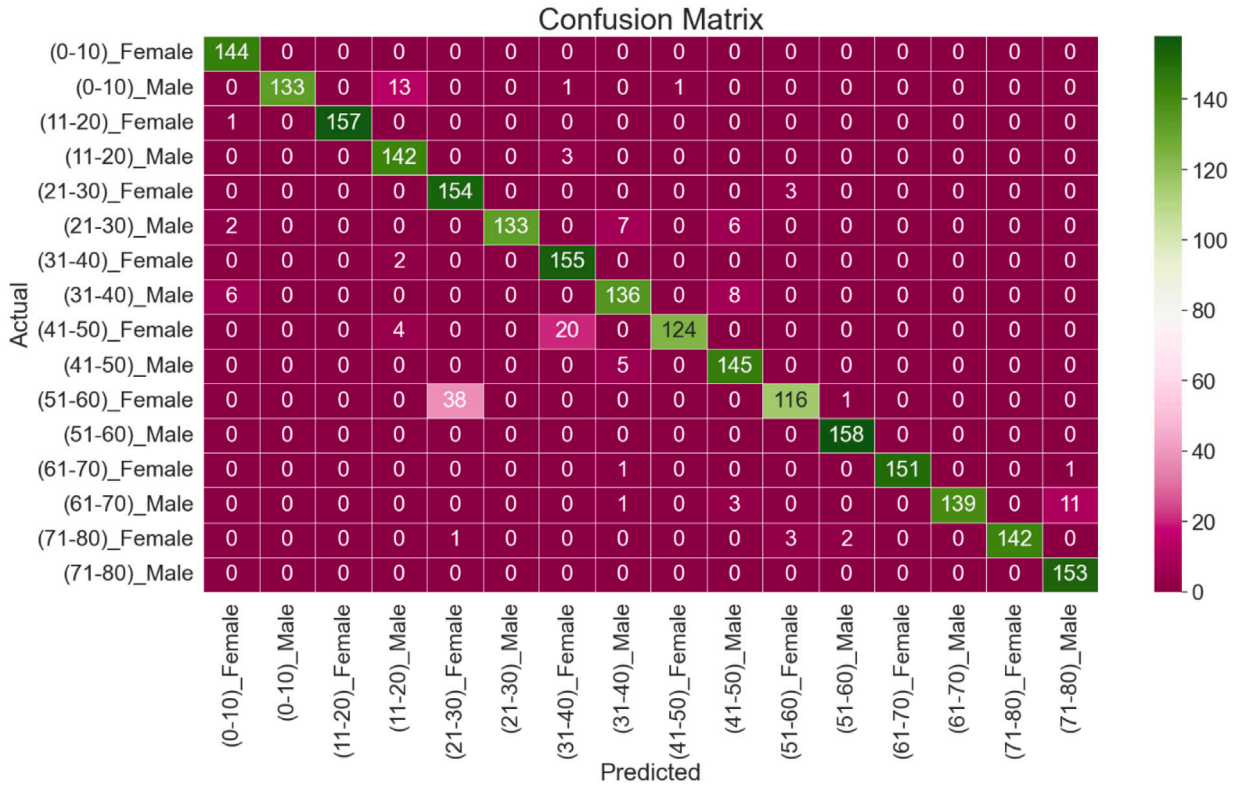


(g) ResNet152V2

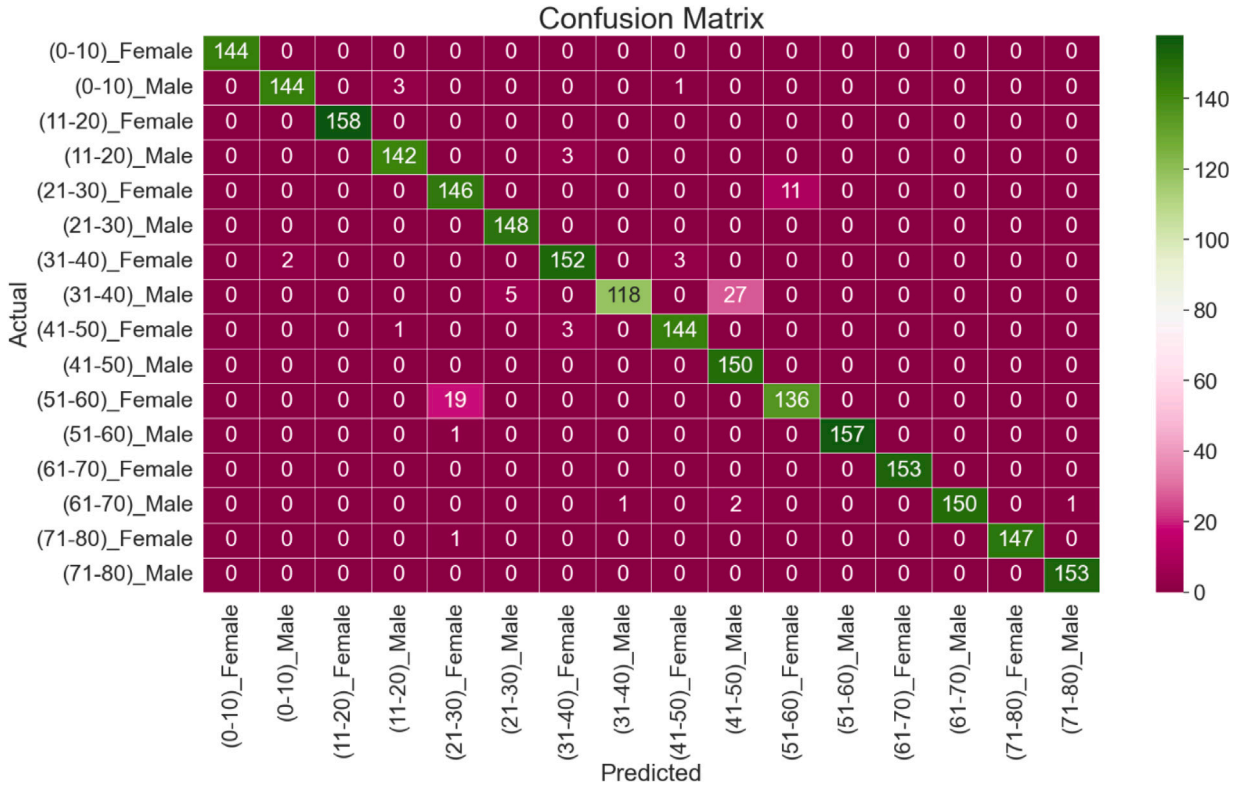


(h) InceptionV3

Fig. 4. (*continued*).

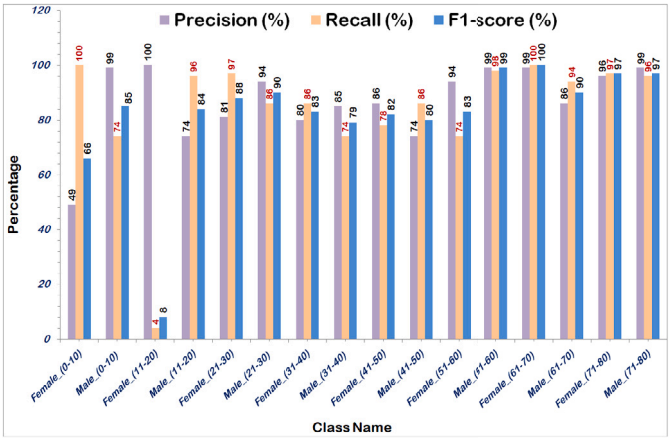


(i) Xception

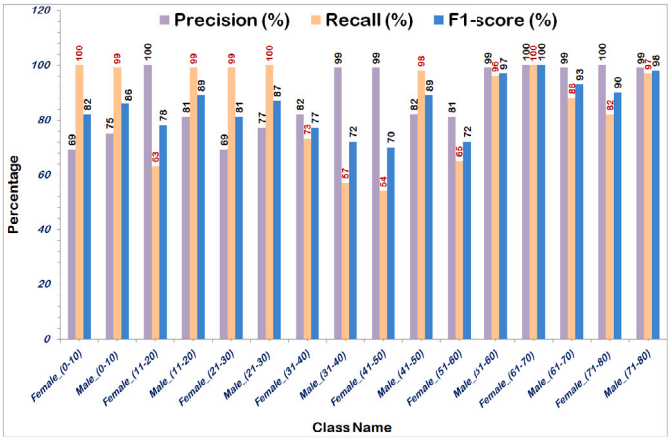


(j) CMNV2 (Proposed)

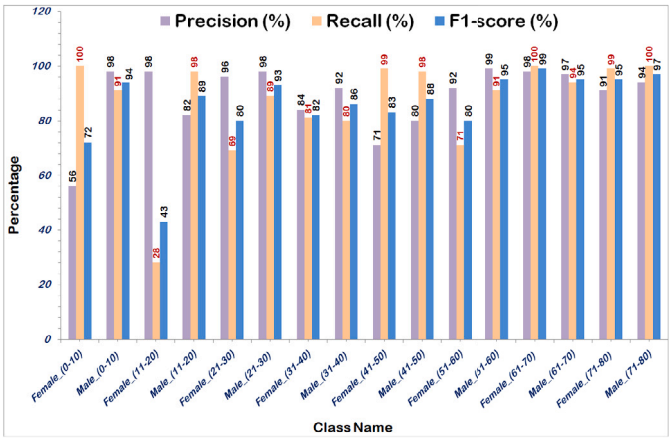
Fig. 4. (*continued*).



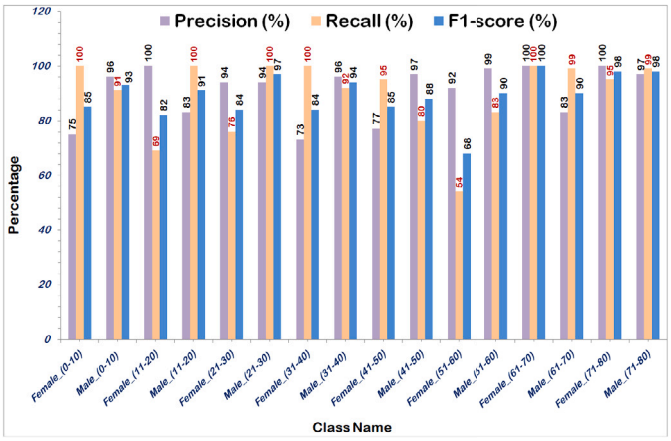
(a) MobileNet



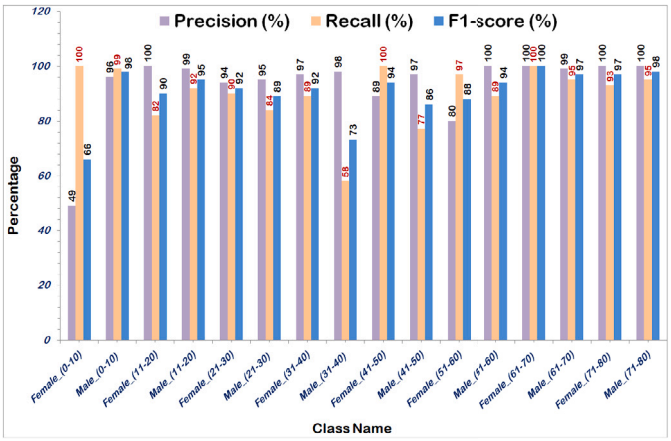
(b) MobileNetV2



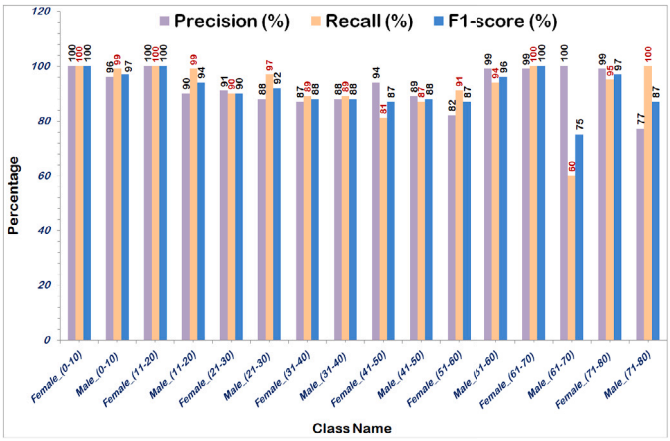
(c) ResNet101



(d) DenseNet169



(e) DenseNet201



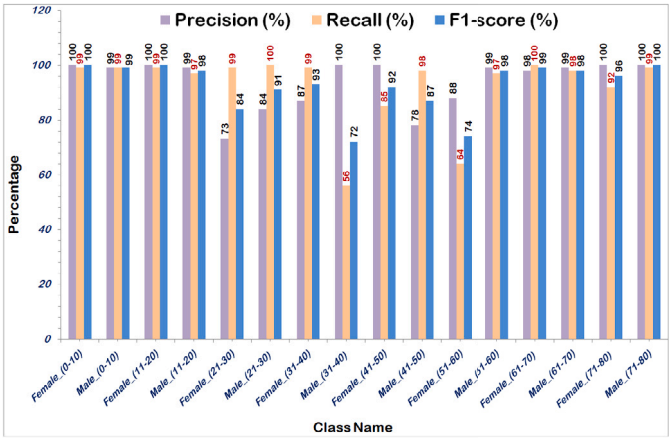
(f) EfficientNetB0

Fig. 5. Comparison graphs statistical analysis of all 10 different models.

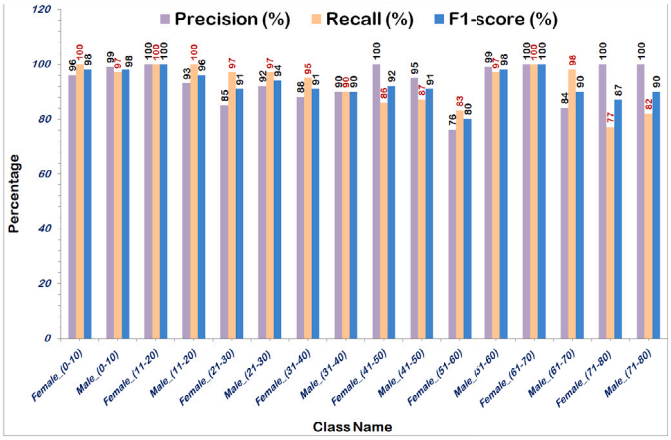
efficiency of 92.91% accuracy. Xception model, the total layers are 135, trainable layers are 135, and non-trainable layers are 0, but with a performance efficiency of 94.06% accuracy.

Ultimately, regarding the performance and the computational efficiency the CMNV2 model's unique layer design is significant for having an optimal balance of learnable and non-learnable layers; out of 165 layers, 8 have been trained, while the remaining layers 157 are used for feature extraction without training using a transfer learning approach. over non-trainable layers, the model maintains a robust and stable model while acquiring valuable features over training, due to the careful layer distribution.

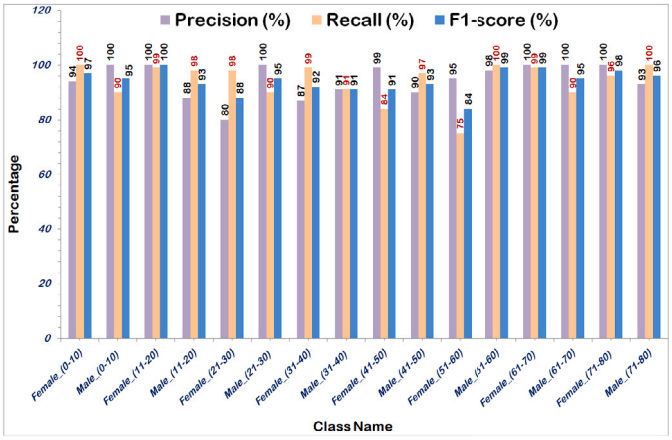
In terms of architectural parameters and run-time efficiency, our proposed CMNV2 model significantly outperforms the remaining 9 standard models. Table 5, contains an in-depth explanation of the run-time performance metrics for each model, total parameters, trained parameters and non-trained parameters, which are presented in Fig. 6(a) to Fig. 6(d). The total trainable parameter values of our CMNV2 approach appear to be the lowest in the category, indicating its strong architectural design. When compared to the other 9 models.



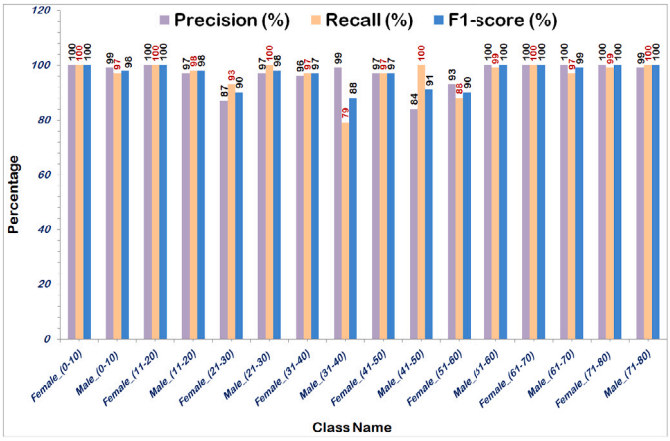
(g) ResNet152V2



(h) InceptionV3



(i) Xception



(j) CMNV2 (Proposed)

Fig. 5. (*continued*).

Table 5
Comparison of parameters and run-time analysis of CMNV2 with different models.

Sl.No.	Name of the model	Parameters in total	Parameters trained	Parameters non-trained	Overall run time (seconds)
1	MobileNet	32,52,432	32,25,424	27,008	47,304
2	MobileNetV2	22,87,440	22,46,928	40,512	40,248
3	ResNet101	4,27,05,296	4,25,89,712	1,15,584	2,58,768
4	DenseNet169	1,26,81,168	1,25,14,448	1,66,720	74,088
5	DenseNet201	1,83,66,160	1,81,27,504	2,38,656	1,37,232
6	EfficientNetB0	40,79,027	40,30,604	48,423	61,272
7	ResNet152V2	5,83,78,768	5,82,24,784	1,53,984	3,37,320
8	InceptionV3	2,18,49,904	2,18,05,232	44,672	1,79,352
9	Xception	2,09,08,600	2,08,43,832	64,768	1,51,344
10	**CMNV2 (Proposed)**	**31,54,896**	**8,90,512**	**22,64,384**	**13,752**

Table 6
Performance analysis comparison of CMNV2 with 10 different models.

Sl.no.	Model name	Accuracy (%)	Error rate (%)
1	MobileNet	83.59	16.41
2	MobileNetV2	85.53	14.47
3	ResNet101	86.40	13.60
4	DenseNet169	89.49	10.51
5	DenseNet201	90.02	9.98
6	EfficientNetB0	91.71	8.29
7	ResNet152V2	92.75	7.25
8	InceptionV3	92.91	7.09
9	Xception	94.06	5.94
10	**CMNV2 (Proposed)**	**96.54**	**3.46**

The CMNV2 approach calculation and learning times are quicker and it can efficiently number of parameters, because of its improved design and excellent computational efficiency, CMNV2 is the best model. As a result, CMNV2 is a perfect approach for many different kinds of MFC, MFAI and MFGI applications. Another two regions where CMNV2 outperforms other models are accuracy and error rate efficiency.

As demonstrated in Table 6 and displayed in Fig. 7, the proposed CMNV2 model outperforms 9 different methods in terms of accuracy and error rate. The accuracy of a model is a key indication of its predictive effectiveness and CMNV2 has an exceptionally high accuracy, guaranteeing it can create accurate predictions. When compared to the other 9 standard techniques, our CMNV2 methodology has an exceptionally low error rate, suggesting few occurrences of wrong classification. CMNV2 outperforms its other 9 competing models due

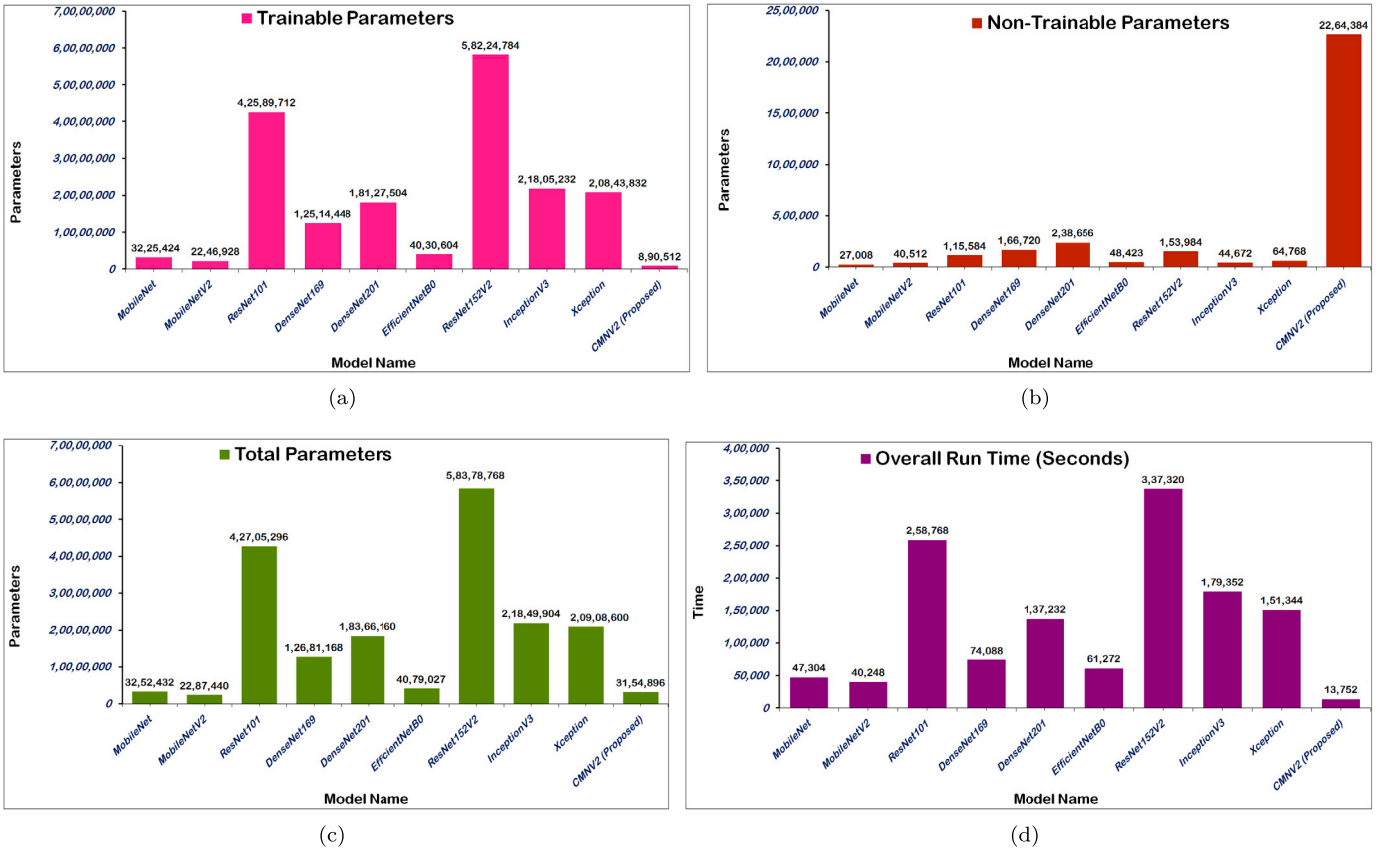


Fig. 6. Comparison graphs analysis of all 10 different models.

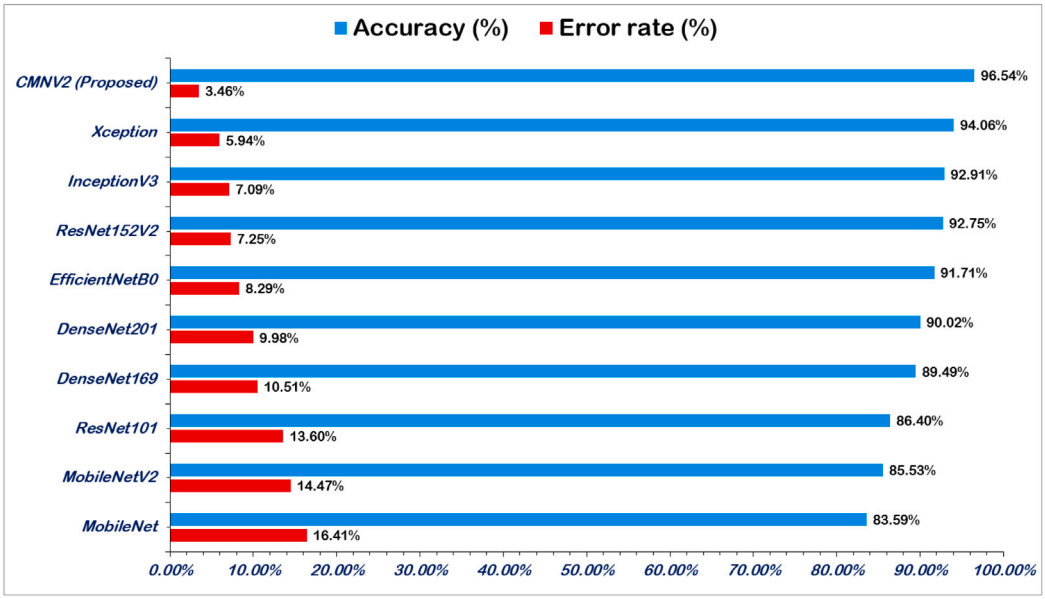


Fig. 7. Accuracy and Error rate comparison graph of all 10 different models.

to its great accuracy and low error rate. High accuracy and low error rates across conditions demonstrate how robust and dependable the suggested CMNV2 model is, making it an excellent alternative for researchers who require accuracy and dependability.

The output results of our proposed CMNV2 model, as shown in Fig. 8, give an in-depth investigation of how effectively it can identify the age and gender of masked faces across 16 distinct classes on photo images. Each of these classes indicates an individual group of circumstances, each with its own set of difficulties. The model's accuracy in identifying the age and gender of an individual in each class who is wearing a mask has been evaluated here. The model's success was dependent on its ability to identify the age and gender of the faces while

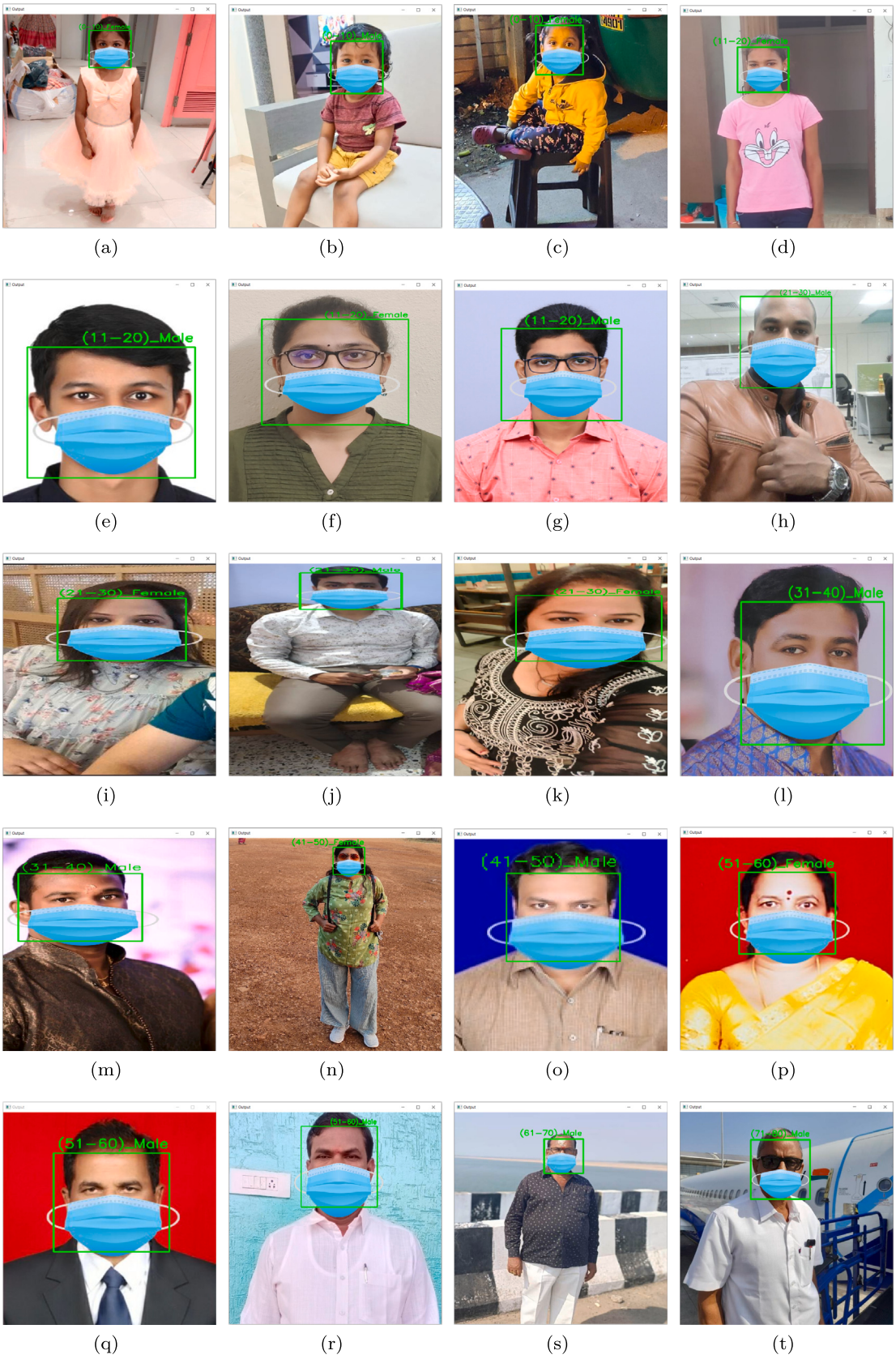


Fig. 8. Masked face age and gender identification of CMNV2 on photo images.

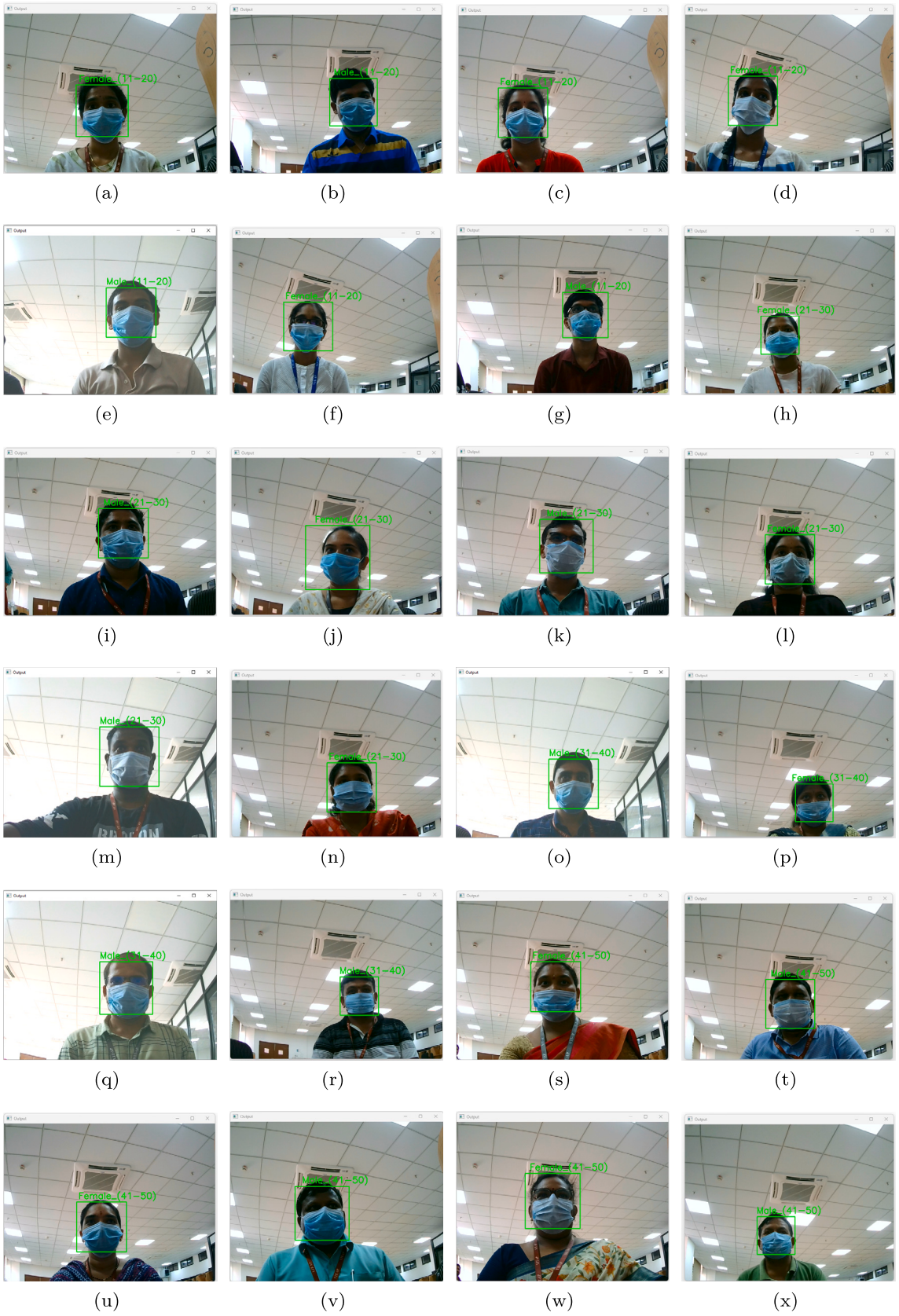


Fig. 9. Masked face age and gender identification of CMNV2 on real-time video images.

wearing masks in a variety of environments and lighting conditions. As seen by its performance throughout these 16 classes, CMNV2 proved a capacity for dependable masked face age and gender identification in a complex context as a consequence of its adaptability and sturdiness. From Fig. 8(a) to Fig. 8(t), a bounding box of a particular green color surrounding the faces accurately shows the identification of masked

faces, with a green label of their people's age and gender names shown on top of each box. The green box shows the exact position of the identified face. At the same time, the green label shown above the bounding box clearly shows the model's prediction, indicating that the model correctly identified the presence of a mask on each face. This study becomes important when accurate identification of people wearing masks is essential as it is in security, healthcare and public health.

In our study, 24 participants voluntarily participated in testing the performance of our novel CMNV2 technique for real-time video image processing and prediction, including both B.Tech students and Ph.D. scholars are shown in Fig. 9. We took measures to provide for the security and comfort of the participants during the testing sessions by providing masks. Each participant participated in the evaluation of the proposed CMNV2 technique while wearing masks for privacy. Our CMNV2 model was used to evaluate real-time video frames to analyze how effective it was in producing predictions efficiently. The participant's collaborative efforts were critical in testing the CMNV2 technique in real-world functionality and robustness in the context of video image processing. Their passion and understanding helped to improve and ensure the successful implementation of our approach, increased technological predictability and encouraged an interesting exchange of ideas and discoveries, advancing in computer vision. These students and scholars have helped us improve our study and push the limits of computer vision in improving our research and testing the boundaries of what is achievable in the field of MFAI and MFGI as displayed in Fig. 9.

In the proposed work each frame is processed with the CMNV2 model on real-time video at 30 frames per second (FPS) is supported by many systems, devices and broadcasting standards. More power is required for streaming or storing at 60 frames per second or higher. Performance is quicker at 30 frames per second with no loss of slow execution. 30 FPS is also often employed since it provides an acceptable combination of a smooth pace and the intense visual impact that can be needed. When compared to greater frame rates, 30 FPS videos are less in size. This can prove advantageous for mobile devices and other platforms with limited storage capacity. A frame rate of 30 FPS helps reduce flickering and improve visual consistency.

The training and testing performance of our proposed CMNV2 model is presented in Fig. 10(a) and 10(b), which indicates training and testing accuracy and loss. The model gradually acquires the ability to identify the age and gender of masked faces, as shown in Fig. 10(a), which displays training accuracy with the 'purple' color and testing accuracy using the 'orange' color. The model's identification capacity is shown by the training and testing accuracy curve.

Fig. 10(b), represents training and testing loss to demonstrate how the model reduces during training failures. A decreasing training loss represented by a 'purple' color curve and a decreasing testing loss represented by an 'orange' color curve show the model is becoming better at making correct predictions. These convergence curves demonstrate that the CMNV2 model has been trained to identify the age and gender of masked faces with low prediction errors. These graphs are critical for evaluating training. These guarantees that the model learns and generalizes without becoming too dependent on training data. They also give valuable data into the model's performance, allowing the CMNV2 model to be modified and optimized for greater accuracy and efficiency in MFAI and MFGI tasks. The region of convergence-area under the curve (ROC-AUC) curve for predicting 16 classes of MFAGI using the CMNV2 model demonstrates good multi-class classification. As seen in Fig. 11, the value is larger than 0.5, indicating that the model outperforms random chance in differentiating between the 16 different classes.

Each of the 16 classes has a unique, ascending curve, demonstrating the model's accurate categorization within masked face classes. This curve balances sensitivity, which is a measure of a model's ability to identify positive cases accurately (true positives) out of all real positive occurrences. Furthermore, specificity is a measure of a model's ability to accurately identify negative cases (true negatives) among all real negative occurrences. These two provide accurate predictions across all classes while being uniform and consistent, exhibiting the model's stability and dependability. The best ROC-AUC curve demonstrates the CMNV2 model's remarkable performance in detailed, multi-class masked face age and gender identification tasks, which makes it applicable in a wide range of applications in the real world.

6. Conclusion

The results of our study on a CMNV2 model for masked face classification (MFC), masked face age identification (MFAI), and masked face gender identification (MFGI) are presented in this research paper. Collecting a dataset of unmasked faces from Kaggle allowed us to evaluate the proposed model's efficiency. To train the facial database for MFC, MFAI, and MFGI, artificial face masking data techniques were used to create a masked face dataset of 12,160 images. For masked face classification of photos utilizing facial data, we started with traditional MobileNetV2, a network with a pre-trained model, employed as a transfer learning approach then fine-tuned with the deep neural network by adding 8 different layers. Making effective use of transfer learning to train the CAFFE model's deep learning features for more quickly and improve its overall performance. Following are the models: MobileNet, MobileNetV2, ResNet101, DenseNet169, DenseNet201, EfficientNetB0, ResNet152V2, InceptionV3, Xception, and our novel CMNV2 model. Used this dataset of 12,160 images with 16 classes to train 10 different models. The testing accuracy of all of these 10 models ranged from 83.59% to 96.54%.

Once data was classified, all of the models correctly and incorrectly identified the age and gender of masked faces, however, the accuracy obtained differed significantly among models. The investigation of 10 different models for masked face age and gender identification (MFAGI), using the artificially created dataset shows that the proposed CMNV2 outperforms the other 9 models. Our CMNV2 model consistently outperformed the other models, with an accuracy of 96.54%, an error rate of 3.46% with significantly fewer training parameters, and a much less computation time. Based on these results, the proposed CMNV2 is clearly the most successful approach for addressing the challenges of MFAGI. The technology that is suggested is crucial because it can identify people with face masks faster and more accurately than the human eye giving good dependability.

In the presence of a mask, the suggested CMNV2 method increases the prediction of mask identification of faces. The proposed CMNV2 technique can help in smart access systems and intelligent attendance systems in institutions of higher learning, healthcare facilities, and other business sectors. Regarding the shortage of masked face datasets, created a dataset containing 12,160 images by artificial masking technique. We believe that the dataset made publicly available on the Kaggle platform will be very useful to researchers working in this field as well as programs that depend on masked face age identification and masked face gender identification.

Limitations: Our research realizes an accuracy difference between still photo images and real-time video situations. While the model works well on static photos, it has challenges when applied to moving environments, particularly real-time video. Such a limitation indicates that the model's performance is affected by factors such as sudden motion, geographical variations and changes in lighting conditions. In real-time video prediction, a particular restriction is revealed concerning decreasing accuracy during sudden movements of participants. This is an important issue in video-based applications, because the quick movements may add unpredictability and limit the model's ability to predict effectively.

Future Work: Future work is likely to focus on improving the model to enhance accuracy in real-time video settings, particularly when there

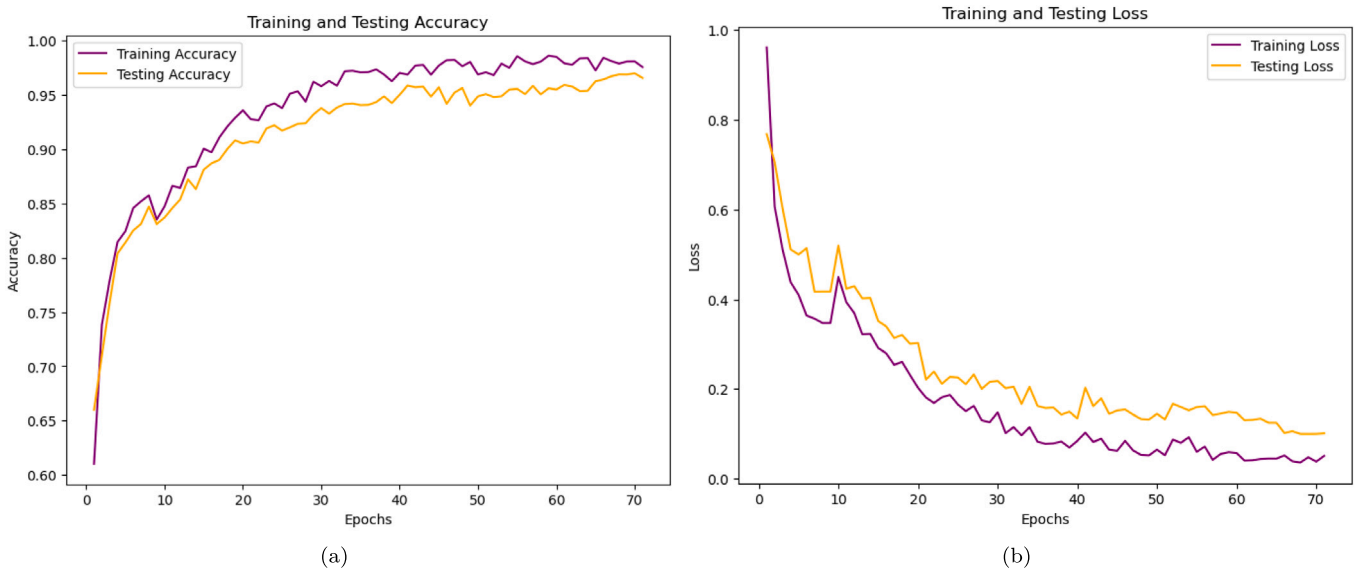


Fig. 10. Graph of training and testing accuracy and loss of CMNV2 model.

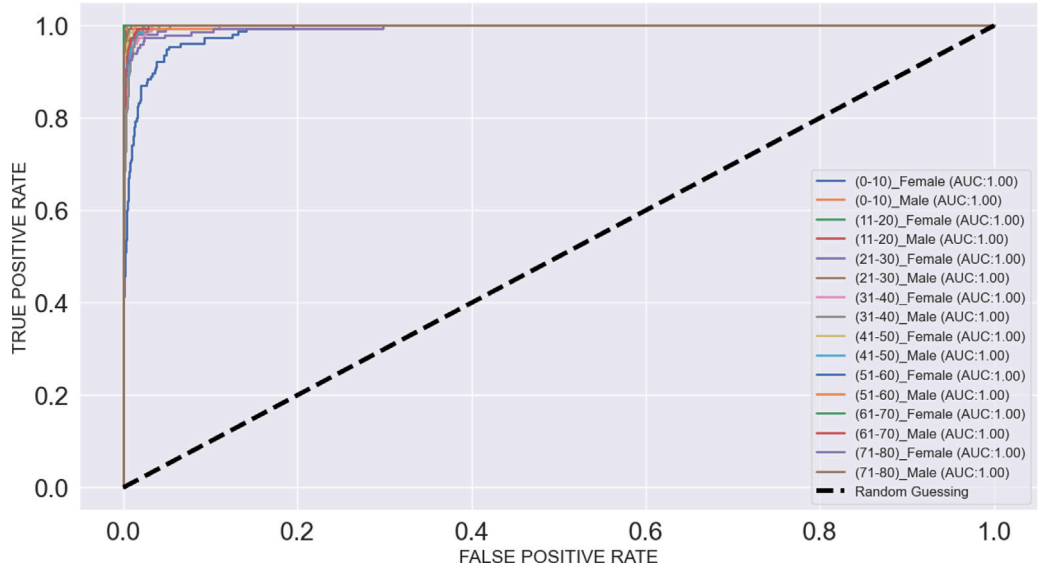


Fig. 11. ROC-AUC curve of CMNV2 model.

are abrupt movements. This might include implementing movement compensation methods and changing the model architecture to better accommodate dynamic changes in input data. Extending the model's capacity to estimate the age and gender of two or more masked faces in real-time video is another area for future investigation for multi-face analysis. This can be especially important in situations such as overcrowded public places or large group gatherings.

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CRediT authorship contribution statement

B.Anil Kumar: Conceptualization, Methodology, Dataset creation, Visualization, Software, Writing – original draft, Validation. **Neeraj Kumar Misra:** Data curation, Project administration, Supervision, Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The artificially masked 12,160 images of the dataset created of masked faces used in this study, made publicly available in the Kaggle https://www.kaggle.com/datasets/banilkumar20phd7071/masked-face-age-and-gender-identify-artificial-masking. Complete coding will be provided on request.

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